Heaviside Composite Optimization Problems

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Take-away lessons from course:

- Heaviside optimization is a novel paradigm in optimization, bridging the continuous and discrete domains.
- The subject is very rich in applications, its rigorous treatment demands advanced optimization background and its efficient solution requires a fusion of continuous and discrete optimization methods.
- Many classical and modern problems in operations research, computational statistics, machine learning, health treatments, and policy learning can be modeled by this framework, with significant benefits and advancements.
- There remain many challenges in theory and algorithms, awaiting to be developed and resolved by the junior talents.

10 chapters from a monograph in progress

J. Liu and J.S. Pang. *Heaviside Composite Optimization Problem: Applications, Theory, Algorithms, and Numerics.*

Chapter I: Introduction

Chapter II: Paradigm Changes: Sources of Discontinuity

Chapter III: Some Prerequisites

Chapter IV: Solution Analysis: The Nonlinear Programming Approach

Chapter V: Affine Heaviside Composite Problems

Chapter VI: Integer Programming Based Algorithms

Chapter VII: Extended Problems

Chapter VIII: Fractional and Doubly Composite Problems

Chapter IX: Numerical Implementation and Results (Junyi Liu)

Chapter X: Case Studies (Junyi Liu)

Supplementary further reading — the monograph in progress and its theoretical foundations

Monograph (in progress). J. Liu and J.-S. Pang. *Heaviside Composite Optimization: Theory, Algorithms, Applications, and Numerics.* 2025.

Theory — pseudo- and epi-stationarity. Y. Cui, J. Liu, J.-S. Pang. *The Minimization of Piecewise Functions: Pseudo Stationarity.* arXiv:2305.14798 (2023); J. Convex Anal. 30(3):793–834.

S. Han, Y. Cui, J.-S. Pang. *Analysis of a Class of Minimization Problems Lacking Lower Semicontinuity.* Math. Oper. Res. 50(3):2175–2198 (2025).

Supplementary further reading — progressive-IP algorithms, applications, and an overview talk

Progressive (M)IP algorithms & applications. Y. Fang, J. Liu, J.-S. Pang. *Classification and Treatment Learning with Constraints via Composite Heaviside Optimization: a Progressive MIP Method.* arXiv:2401.01565 (2024).

K. Zheng, J. Liu, Y. Wang, J.-S. Pang. *Solving Constrained Affine Heaviside Composite Optimization Problems by a Progressive IP Approach.* arXiv:2605.06020 (2026).

Jingren Liu, H. Qin, Junyi Liu, M. C. Chou, J.-S. Pang. *Offline Policy Learning with Weight Clipping and Heaviside Composite Optimization.* arXiv:2601.12117 (2026).

Overview talk. J.-S. Pang. *Heaviside Composite Optimization: A New Paradigm in Optimization.*

Chapter I

Introduction

A bird's eye-view of post World-War-II optimization till now:

- From linear programming to nonlinear and integer programming.
- From nonlinear programming to stochastic programming and modern-day convex programming.
- From differentiability to nondifferentiability.
- From nondifferentiability to semicontinuity.
- From convexity to nonconvexity.

Y. Cui and J. S. Pang, *Modern Nonconvex Nondifferentiable Optimization*. SIAM, MOS–SIAM Series on Optimization (December 2021).

Apart from further developments of the present, what is the future??

From semicontinuity to lack thereof $\Rightarrow$ discontinuous optimization.

From closedness of constraints to non-closedness.

The General Heaviside Composite Optimization Problem (G-HSCOP)

$$\underset{\mathbf{x}\in P}{\textbf{maximize}} \quad \Phi(\mathbf{x}) \triangleq c(\mathbf{x}) + \sum_{k=1}^{K_0} \psi_{0k}(\mathbf{x}) \quad \underbrace{\mathbb{1}_{[0,\infty)}(\phi_{0k}(\mathbf{x}))}_{\text{composite indicator fnc.}}$$

$$\textbf{subject to} \quad \mathbf{A}_{i\bullet}\mathbf{x} + \sum_{k=1}^{K_i} \psi_{ik}(\mathbf{x}) \mathbb{1}_{[0,\infty)}(\phi_{ik}(\mathbf{x})) \geq \eta_i, \qquad i = 1, \ldots, I.$$

- P is a friendly set, c is a friendly function, e.g., affine or quadratic; friendly = amenable to well-known treatments, and $\mathbf{A}_{i\bullet}$ is the i-th row of a matrix $\mathbf{A} \in \mathbb{R}^{I\times n}$.
- ψ_{ik} and ϕ_{ik} are functions whose properties highlight the challenges;
- most importantly, $\mathbb{1}_{[0,\infty)}$ is the "closed" Heaviside function given by

$$\mathbb{1}_{[0,\infty)}(s) \triangleq \begin{cases} 1, & \text{if } s \in [0,\infty), \\ 0, & \text{otherwise,} \end{cases} \qquad \text{(upper semi-continuous).}$$

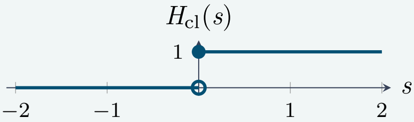
The General Heaviside Composite Optimization Problem
(G-HSCOP)

Supplementary visualization: closed vs. open Heaviside thresholds

The Heaviside indicator turns a scalar condition into a binary switch. The bracket at $s = 0$ determines both activation at equality and the direction of semicontinuity.

Closed threshold: $[0, \infty)$
Upper, not lower, semicontinuous; favorable for $\max$.

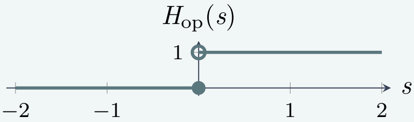
$$H_{\mathrm{cl}}(s) \triangleq \mathbb{1}_{[0,\infty)}(s) = \begin{cases} 0, & s < 0, \\ 1, & s \geq 0. \end{cases}$$



$H_{\mathrm{cl}}(0) = 1$; equality activates the term.

Open threshold: $(0, \infty)$
Lower, not upper, semicontinuous; favorable for $\min$.

$$H_{\mathrm{op}}(s) \triangleq \mathbb{1}_{(0,\infty)}(s) = \begin{cases} 0, & s \leq 0, \\ 1, & s > 0. \end{cases}$$



$H_{\mathrm{op}}(0) = 0$; strict inequality is required. In particular, $|t|_0 = H_{\mathrm{op}}(|t|)$.

Thus $\psi(\mathbf{x}) H_{\mathrm{cl}}(\phi(\mathbf{x}))$ equals $\psi(\mathbf{x})$ when $\phi(\mathbf{x}) \geq 0$, and equals 0 otherwise.

Conditional Functions and Sparsity

The product $f(\mathbf{x})\mathbb{1}_{[0,\infty)}(g(\mathbf{x}))$ is a conditional function; i.e., the primary function $f(\mathbf{x})$ is activated only if the auxiliary condition $g(\mathbf{x}) \geq 0$ holds.

An immediate example: the sparsity (counting) function

$$|t|_0 \triangleq \begin{cases} 1, & \text{if } t \neq 0, \\ 0, & \text{if } t = 0, \end{cases} = \mathbb{1}_{(0,\infty)}(|t|) = 1 - \mathbb{1}_{[0,\infty)}(-|t|),$$

illustrates that the function inside the Heaviside function may not be differentiable. This function extends to any step function with a finite number of (discontinuous) jumps.

An affine sparsity constraint

$$\sum_{j=1}^{n} a_j |x_j|_0 \leq b, \qquad \text{with } a_j \text{ of mixed signs,}$$

is a non-trivial extension of the cardinality constraint $\sum_{j=1}^{n} |x_j|_0 \leq K$, whose mixing coefficients are all positive.

Supplementary note. Positive coefficients preserve lower semicontinuity. With mixed-sign a_j, upward and downward jumps can coexist, so $\sum_{j=1}^{n} a_j |x_j|_0$ may be neither lower nor upper semicontinuous.

References

Methodology

- Y. Cui, J. Liu, and J.S. Pang. On the minimization of piecewise functions: Pseudo stationarity. *Journal of Convex Analysis* 30(3): 793–834 (2023).
- S. Han, Y. Cui, and J.S. Pang. Analysis of a class of minimization problems lacking lower semicontinuity. *Mathematics of Operations Research* 50(3): 2175–2198 (2025-8).
- K. Zheng, J. Liu, R. Wang, and J.S. Pang. Solving constrained affine Heaviside composite optimization problems by a progressive IP approach. Manuscript (April 2026).

Applications

- Z. Qi, Y. Cui, Y. Liu, and J.S. Pang. Estimation of individualized decision rules based on optimized covariate-dependent equivalent of random outcomes. *SIAM Journal on Optimization* 29(3): 2337–2362 (2019).
- Y. Fang, J. Liu, and J.S. Pang. Treatment learning with Gini constraints by Heaviside composite optimization and a progressive method. *Computational Optimization and Applications* 92(2): 471–513 (2025-11).

Distinguished (and challenging) features:

- Discontinuity of the Heaviside function and nondifferentiability of the other functions:
  - — lack of semicontinuity in the objective function
  - — non-closedness of the feasible set
  - — does convexity/concavity have a role to play?
- Composition of the highlighted features:
  - — what is a reasonable concept of a solution?
  - — algorithmically, how to compute such a solution?
- Fusion of continuous variables and functions with discrete structure of the Heaviside function; thus natural to consider
  - — a marriage between the continuous and discrete world
  - — how to facilitate the marriage? will it be fruitful?

Chapter II

Prerequisites, a second course in optimization

Prerequisites, a second course in optimization

- A graduate course in linear, differentiable, and integer programming
- **Bouligand differentiability**: broad and most basic for nondifferentiable functions
- **Stationarity conditions** of nondifferentiable programs
- **Piecewise affine functions**: definition and properties
- **Difference-of-convex programs**: criticality vs. stationarity; the difference-of-convex algorithm and its refinement
- **Complementarity constraints**: modeling breadth and connections to binary programming
- **Fractional and composite programs**: Dinkelbach's algorithm and its extension.

Supplementary note. $0 \leq a \perp b \geq 0$ is an or-gate — $a_i = 0$ *or* $b_i = 0$ — i.e. if-then logic written in continuous variables.

Bouligand differentiability

A function $f : \mathcal{O} \subseteq \mathbb{R}^n \to \mathbb{R}$, where $\mathcal{O}$ is an open set in $\mathbb{R}^n$ is Bouligand differentiable at $\bar{\mathbf{x}} \in \mathcal{O}$ if it is locally Lipschitz continuous near $\bar{\mathbf{x}}$ and directionally differentiable there with the (finite) directional derivative along the direction $\mathbf{v} \in \mathbb{R}^n$ defined by:

$$f'(\bar{\mathbf{x}}; \mathbf{v}) \triangleq \lim_{\tau \downarrow 0} \frac{f(\bar{\mathbf{x}} + \tau \mathbf{v}) - f(\bar{\mathbf{x}})}{\tau}.$$

As a function of the direction, $f'(\bar{\mathbf{x}}; \bullet)$ is positively homogeneous and globally Lipschitz continuous on $\mathbb{R}^n$ with the same Lipschitz modulus as that of f near $\bar{\mathbf{x}}$.

Calculus rules hold for the directional derivative; these are the sum, multiplication, division, and composition rules; in particular,

$$(f \circ g)'(\bar{\mathbf{x}}; \mathbf{v}) = f'\big(g(\bar{\mathbf{x}}); g'(\bar{\mathbf{x}}; \mathbf{v})\big),$$

Remark: the partial differentiation rule may fail;

$$f'(\bar{\mathbf{x}}; \mathbf{v}) \neq \sum_{i=1}^{n} f'(\bar{\mathbf{x}}; v_i \mathbf{e}^i), \qquad \text{where } \mathbf{e}^i = i\text{-th coordinate vector}$$

Supplementary note. Convex functions are directionally differentiable.

Piecewise affine functions

A continuous function $f\colon \mathbb{R}^n \to \mathbb{R}^m$ is piecewise affine if there exists a partition of $\mathbb{R}^n$ into a finite number of polyhedra, on each of which, f is an affine function.

A piecewise affine function is Bouligand differentiable everywhere. Most interested in $m = 1$.

A continuous function $f\colon \mathbb{R}^n \to \mathbb{R}^m$ is piecewise affine (PA) if and only if

$$f(\mathbf{x}) = \underbrace{\max_{1 \le k \le K} \left[(\mathbf{a}^k)^\top \mathbf{x} + \alpha_k \right]}_{\text{convex in } \mathbf{x}} - \underbrace{\max_{1 \le \ell \le L} \left[(\mathbf{b}^\ell)^\top \mathbf{x} + \beta_\ell \right]}_{\text{convex in } \mathbf{x}}, \qquad \mathbf{x} \in \mathbb{R}^n$$

for some constant vectors $\mathbf{a}^k$ and $\mathbf{b}^\ell$ and scalars α_k and β_ℓ. Hence, if f is PA, then for every $\bar{\mathbf{x}} \in \mathbb{R}^n$,

$$f(\mathbf{x}) = f(\bar{\mathbf{x}}) + f'(\bar{\mathbf{x}}; \mathbf{x} - \bar{\mathbf{x}}), \qquad \forall\, \mathbf{x} \text{ sufficiently near } \bar{\mathbf{x}},$$

Stationarity conditions

Consider an abstract optimization problem

$$\operatorname*{\textbf{minimize}}_{\mathbf{x}\in X}\ f(\mathbf{x}), \qquad f \text{ is Bouligand differentiable.}$$

A feasible solution $\bar{\mathbf{x}} \in X$ is a Bouligand stationary solution if

$$\boxed{f'(\bar{\mathbf{x}};\mathbf{v}) \geq 0 \ \text{ for all tangent vectors } \mathbf{v} \in \mathcal{T}(X;\bar{\mathbf{x}})}, \quad \text{where}$$

$\mathcal{T}(X;\bar{\mathbf{x}}) \triangleq$ tangent cone of X at $\bar{\mathbf{x}}$

$$\left\{ \mathbf{v} \in \mathbb{R}^n \;\middle|\; \begin{array}{l} \mathbf{v} = \displaystyle\lim_{k\to\infty} \frac{\mathbf{x}^k - \bar{\mathbf{x}}}{\tau_k} \text{ for some} \\ \text{sequences } \{\mathbf{x}^k\} \subset X \text{ converging} \\ \text{to } \bar{\mathbf{x}} \text{ and } \{\tau_k\} \downarrow 0 \end{array} \right\}$$

Two questions: How to describe the tangent vectors? What if f is not directionally differentiable?

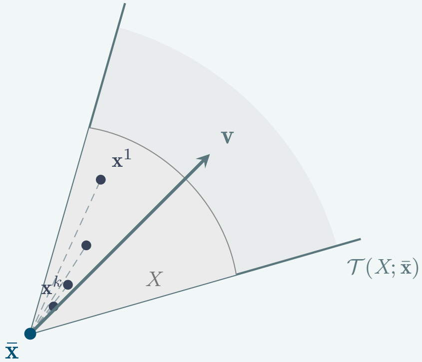
A feasible solution $\bar{\mathbf{x}} \in X$ is an epi-stationary solution if the pair $(\bar{t}, \bar{\mathbf{x}})$, where $\bar{t} \triangleq f(\bar{\mathbf{x}})$, is a Bouligand stationary solution of the lifted problem:

$$\operatorname*{\textbf{minimize}}_{(t,\mathbf{x})\in\mathbb{R}\times X}\ t \quad \textbf{subject to} \ \ t \geq f(\mathbf{x}).$$

Note: if f is Bouligand differentiable, then Bouligand stationarity $\Leftrightarrow$ epi-stationarity.

Stationarity conditions

Supplementary visualization: the tangent cone as a limit of secant directions



Each feasible $\mathbf{x}^k \in X$ near $\bar{\mathbf{x}}$ gives a secant direction $(\mathbf{x}^k - \bar{\mathbf{x}})/\tau_k$. As $\mathbf{x}^k \to \bar{\mathbf{x}}$ the secant slopes converge to a tangent vector $\mathbf{v}$.

The tangent cone $\mathcal{T}(X;\bar{\mathbf{x}})$ (shaded) collects *all* such limiting directions.

Supplementary note. If $\bar{\mathbf{x}}$ is a local minimizer, it must be Bouligand stationary. The condition is necessary, not sufficient — it only identifies $\bar{\mathbf{x}}$ as a candidate solution.

Two set-theoretic properties: A set $X \subseteq \mathbb{R}^n$ is said to be

- locally convex-like near $\bar{\mathbf{x}} \in X$ if there exists a neighborhood $\mathcal{N}$ of $\bar{\mathbf{x}}$ such that $X \cap \mathcal{N} \subseteq \bar{\mathbf{x}} + \mathcal{T}(\bar{\mathbf{x}}; X)$;
- locally star-shaped near $\bar{\mathbf{x}} \in X$ if for any $\mathbf{x} \in X$ sufficiently close to $\bar{\mathbf{x}}$, there exists a scalar $\bar{\tau} > 0$ (dependent on $\mathbf{x}$) such that $\bar{\mathbf{x}} + \tau(\mathbf{x} - \bar{\mathbf{x}}) \in X$ for all $\tau \in [0, \bar{\tau}]$.

Clearly, convex $\Rightarrow$ locally star-shaped $\Rightarrow$ locally convex-like.

- Importantly, if f is Bouligand differentiable and satisfies

$$f(\mathbf{x}) \geq f(\bar{\mathbf{x}}) + f'(\bar{\mathbf{x}}; \mathbf{x} - \bar{\mathbf{x}}), \qquad \forall\, \mathbf{x} \text{ sufficiently near } \bar{\mathbf{x}}, \tag{1}$$

  then $\bar{\mathbf{x}}$ is Bouligand stationary on a locally convex-like set X if and only if $\bar{\mathbf{x}}$ is a local minimizer of f on X.

- Appropriate compositions of convex and PA functions satisfy (1):

$$f = \varphi \circ \Theta \circ \psi, \qquad \Theta \triangleq (\theta_\ell)_{\ell=1}^{L},$$

  where $\varphi : \mathbb{R}^L \to \mathbb{R}$ is PA and isotone, $\theta_\ell : \mathbb{R}^m \to \mathbb{R}$ for $\ell = 1, \ldots, L$ is convex, and $\psi : \mathbb{R}^n \to \mathbb{R}^m$ is PA.

Chapter III

Paradigm changes: Sources of discontinuity

Modeling breadth (OR, statistics, ML, and policy learning)

- Set-up costs— classical in operations research, today's term is sparsity
- Structured variable/constraint selection; on-off constraints
- Conditional stochastic functionals — probabilities and expectations
- Statistical models — conformal regression; change-point detection; robust trimming; and more
- Classification — score-based binary classification; multiclass classification; advanced measures; decision trees
- Clustering: refreshed and extended — mixed-norm: extending traditional K-means and K-clustering; bicriteria (errors versus similarities)
- Univariate and vector quantization
- Policy learning and causal inference — piecewise affine policy for stochastic optimization; causal inference for individualized treatment learning; policy learning with optimal weight clipping

A host of instances

from classical to modern to future

Hierarchical variable selection

The statistical model with interaction terms: hypothesizing a bilinear relation among the input features $\boldsymbol{\xi} = (\xi_i)_{i=1}^d$

$$m(\mathbf{x}; \boldsymbol{\xi}) = \sum_{i=1}^{d} x_i^{(1)} \xi_i + \sum_{1 \le i < j \le d} x_{ij}^{(2)} \xi_i \xi_j,$$

with the parameter $\mathbf{x}$ consisting of the unknown model coefficients $x_i^{(1)}$ and $x_{ij}^{(2)}$ that obey

- strong hierarchy: an interaction term can be selected only if both of the linear terms are selected, i.e.,
$$\left|x_{ij}^{(2)}\right|_0 \le \min\left\{\left|x_i^{(1)}\right|_0, \left|x_j^{(1)}\right|_0\right\};$$
- weak hierarchy: an interaction term can be selected only if at least one of the corresponding linear terms is selected,
$$\left|x_{ij}^{(2)}\right|_0 \le \left|x_i^{(1)}\right|_0 + \left|x_j^{(1)}\right|_0;$$

both leading to mixed-sign combinations of Heaviside functions.

Conditional stochastic functionals

- Expression of a probability function as an expectation:

$$\mathbb{P}\left[f(\mathbf{x},\boldsymbol{\xi}) \geq 0\right] = \mathbb{E}\,\mathbb{1}_{[0,\infty)}\left(f(\mathbf{x},\boldsymbol{\xi})\right) \xrightarrow[\text{approximation (SAA)}]{\text{sample average}} \frac{1}{N}\sum_{s=1}^{N}\mathbb{1}_{[0,\infty)}\left(f(\mathbf{x},\boldsymbol{\xi}^s)\right);$$

- and by extension, expression of a decision-dependent conditional expectation function:

$$\mathbb{E}\left[\psi(\mathbf{x},\boldsymbol{\xi}) \mid f(\mathbf{x},\boldsymbol{\xi}) \geq 0\right] \triangleq \underbrace{\frac{\mathbb{E}\left[\psi(\mathbf{x},\boldsymbol{\xi})\mathbb{1}_{[0,\infty)}(f(\mathbf{x},\boldsymbol{\xi}))\right]}{\mathbb{P}\left[f(\mathbf{x},\boldsymbol{\xi}) \geq 0\right]}}_{\text{fractional Heaviside function}}.$$

Note the product

$$\psi(\mathbf{x},\boldsymbol{\xi})\,\mathbb{1}_{[0,\infty)}\left(f(\mathbf{x},\boldsymbol{\xi})\right)$$

raising the possibility of losing semicontinuity (in the decision variable $\mathbf{x}$) if the sign of the multiplier function $\psi(\bullet,\boldsymbol{\xi})$ is not consistent with the optimization.

Mixed-sign combinations and piecewise inner functions

- First-order stochastic dominance:

$$\mathbb{P}\big[f_1(\mathbf{x},\boldsymbol{\xi}) \le c_1\big] \le \mathbb{P}\big[f_2(\mathbf{x},\boldsymbol{\xi}) \le c_2\big]$$

- Conjunctive probability constraint:

$$\begin{aligned} b \;&\ge\; \mathbb{P}\big[\psi_1(\mathbf{x},\boldsymbol{\xi}) \ge 0 \;\text{ and }\; \psi_2(\mathbf{x},\boldsymbol{\xi}) < 0\big] \\ &=\; \mathbb{P}\big[\psi_1(\mathbf{x},\boldsymbol{\xi}) \ge 0\big] - \mathbb{P}\big[\min\big(\psi_1(\mathbf{x},\boldsymbol{\xi}),\psi_2(\mathbf{x},\boldsymbol{\xi})\big) \ge 0\big] \end{aligned}$$

- Conditional probability constraint:

$$b \ge \mathbb{P}\big[\psi_1(\mathbf{x},\boldsymbol{\xi}) \ge 0 \;\big|\; \psi_2(\mathbf{x},\boldsymbol{\xi}) \ge 0\big]$$

$$\Updownarrow$$

$$\mathbb{P}\big[\min\big(\psi_1(\mathbf{x},\boldsymbol{\xi}),\psi_2(\mathbf{x},\boldsymbol{\xi})\big) \ge 0\big] - b\,\mathbb{P}\big[\psi_2(\mathbf{x},\boldsymbol{\xi}) \ge 0\big] \le 0.$$

Some statistical models (communicated by Dr. Yao Xie)

- **Change-point detection with sparsity.** Let $\{\mathbf{x}_t\}_{t=1}^T$ be a sequence with $\mathbf{x}_t \in \mathbb{R}^p$ and multiple unknown change point $\{\tau_i\}_{i=1}^I$ with mean shifts $\{\boldsymbol{\Delta}_i\}_{i=1}^I$ to be determined. Let $\boldsymbol{\mu} \in \mathbb{R}^p$ be the baseline mean vector which is unknown. This problem is formulated as follows:

$$\begin{aligned}
\underset{\boldsymbol{\mu},\{\boldsymbol{\Delta}_i\},\{\tau_i\}}{\textbf{minimize}} \quad & \sum_{t=1}^{T} \mathbb{1}_{(0,\infty)}(\tau_1 - t)\big\|\mathbf{x}_t - \boldsymbol{\mu}\big\|_2^2 + \sum_{i=1}^{I}\big\|\boldsymbol{\Delta}_i\big\|_0 \\
& + \sum_{i=1}^{I}\sum_{t=1}^{T} \boxed{\underbrace{\mathbb{1}_{(0,\infty)}(t-\tau_i)\,\mathbb{1}_{[0,\infty)}(\tau_{i+1}-t)}_{\text{expressing } \tau_i < t \le \tau_{i+1}}}\ \big\|\mathbf{x}_t - \boldsymbol{\mu} - \boldsymbol{\Delta}_i\big\|_2^2 \\
\textbf{subject to} \quad & \tau_1 \le \tau_2 \le \cdots \le \tau_I.
\end{aligned}$$

Note the product of an open and a closed Heaviside function.

Supplementary note. Matched brackets let two factors collapse into one, $\mathbb{1}_{[0,\infty)}(a)\,\mathbb{1}_{[0,\infty)}(b) = \mathbb{1}_{[0,\infty)}(\min(a,b))$, and likewise for $\mathbb{1}_{(0,\infty)}$. The mixed pair above does not, since $\{a > 0,\ b \ge 0\}$ is neither open nor closed. Under $\tau_i \le \tau_{i+1}$ it is instead the mixed-sign difference $\mathbb{1}_{[0,\infty)}(\tau_{i+1} - t) - \mathbb{1}_{[0,\infty)}(\tau_i - t)$, with affine inner functions.

Some statistical models (cont.)

- **Trimmed least-squares for robust statistical learning.**

$$\underset{f\in\mathcal{F};\,\gamma\geq 0}{\textbf{minimize}} \quad \sum_{i=1}^{n} \underbrace{\mathbb{1}_{[0,\infty)}\big(\underbrace{\gamma}_{\text{trimming threshold}} - \|\mathbf{y}_i - f(\mathbf{x}_i)\|\big)\,\big\|\mathbf{y}_i - f(\mathbf{x}_i)\big\|_2^2}_{\text{a conditional error}}$$

$$\textbf{subject to} \quad \underbrace{\frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{[0,\infty)}\big(\gamma - \|\mathbf{y}_i - f(\mathbf{x}_i)\|\big) \geq 1-\beta}_{\text{bounding the trimming ratio}}\,.$$

Clustering problems: Classical and extended.

Let $\Xi \triangleq \{\boldsymbol{\xi}^s\}_{s=1}^N$ be a given set of data points in $\mathbb{R}^p$. Let $\mathbf{c} \triangleq \{\mathbf{c}^k\}_{k=1}^K$ be an unknown set of cluster centers in $\mathbb{R}^p$. The criterion for a data point $\boldsymbol{\xi}^s$ to belong to cluster k, written as $s \in k$, is

$$\left\|\boldsymbol{\xi}^s - \mathbf{c}^k\right\| \le \min_{k' \neq k}\left\|\boldsymbol{\xi}^s - \mathbf{c}^{k'}\right\|$$

$$\Leftrightarrow \quad h_s(\mathbf{c}) \triangleq \min_{k' \neq k}\left\|\boldsymbol{\xi}^s - \mathbf{c}^{k'}\right\| - \left\|\boldsymbol{\xi}^s - \mathbf{c}^k\right\| \ge 0.$$

Define the simple indicator

$$z_{sk} \triangleq \mathbb{1}(s \in k) \triangleq \mathbb{1}_{[0,\infty)}\left\{h_s(\mathbf{c})\right\} = \begin{cases} 1 & \text{if } s \in k \\ 0 & \text{if } s \notin k \end{cases}$$

The classical K-means $(p = 2)$ and K-medians $(p = 1)$ problems:

$$\begin{array}{rll} \underset{\mathbf{c};\,\mathbf{z}}{\textbf{minimize}} & \dfrac{1}{N}\displaystyle\sum_{s=1}^{N}\left[\sum_{k=1}^{K} z_{sk}\left\|\boldsymbol{\xi}^s - \mathbf{c}^k\right\|_p^p\right] & \\ \textbf{subject to} & \displaystyle\sum_{k=1}^{K} z_{sk} = 1, \quad \forall\, s = 1, \ldots, N & \text{every sample belongs to at least one cluster} \\ \text{and} & z_{sk} \in \{0,1\}, \quad \forall\,(s,k) \in [N] \times [K]. & \end{array}$$

Clustering problems: Classical and extended. **(cont.)**

Unifying the above two problems is the mixed $\ell_{p,q}$-clustering problem with arbitrary pairs of integers $p, q \geq 1$:

$$\underset{\mathbf{c},\mathbf{z}}{\textbf{minimize}} \quad \underbrace{\frac{1}{N}\sum_{s=1}^{N}\left[\sum_{k=1}^{K} z_{sk}\left\|\boldsymbol{\xi}^s - \mathbf{c}^k\right\|_p^p\right]}_{\text{bi-convex; denoted } \mathrm{WCD}_p(\mathbf{c},\mathbf{z})}$$

ℓ_p-norm in objective

$$\textbf{subject to} \quad \sum_{k=1}^{K} z_{sk} = 1, \quad \forall\, s \in [N] \triangleq \{1, \ldots, N\}$$

and for all $(s, k) \in [N] \times [K]$:

$$\underbrace{z_{sk} \leq \mathbb{1}_{[0,\infty)}\left(\min_{k' \neq k}\left\|\boldsymbol{\xi}^s - \mathbf{c}^{k'}\right\|_q - \left\|\boldsymbol{\xi}^s - \mathbf{c}^k\right\|_q\right)}_{\text{Heaviside constraint with } \ell_q\text{-norm}}$$

$z_{sk} \in \{0, 1\}$, (binary variables)

Note: The Heaviside constraint is not needed in the classical K-means and K-medians problems where $p = q$.

Supplementary note. When $p = q$ the objective alone decides membership. Writing the indicator out separates the cost (ℓ_p) from the rule (ℓ_q), making the rule a free modeling choice.

Chapter IV

Solution analysis

Recall: **The G-HSCOP with feasible set denoted X_{GHS}:**

$$\underset{\mathbf{x}\in P}{\textbf{maximize}} \quad \theta_{\mathrm{GHS}}(\mathbf{x}) \triangleq c(\mathbf{x}) + \sum_{j=1}^{J_0} \psi_{0j}(\mathbf{x}) \mathbb{1}_{[0,\infty)}\big(\phi_{0j}(\mathbf{x})\big)$$

$$\textbf{subject to} \quad \mathbf{A}_{i\bullet}\mathbf{x} + \sum_{j=1}^{J_i} \psi_{ij}(\mathbf{x}) \mathbb{1}_{[0,\infty)}\big(\phi_{ij}(\mathbf{x})\big) \geq \eta_i, \quad i = 1, \ldots, I.$$

Non-attainment of optimal solution. The problem

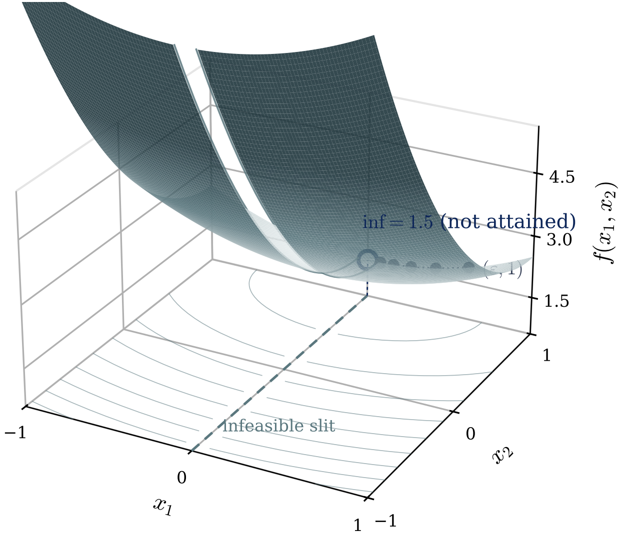
$$\underset{x_1, x_2}{\textbf{minimize}} \quad x_1^2 + 2(1 - x_2)^2 + |x_1|_0 + \tfrac{1}{2}|x_2|_0$$

$$\textbf{subject to} \quad |x_1|_0 \geq |x_2|_0 \ \text{ and } \ -1 \leq x_1, x_2 \leq 1.$$

has an infimum achieved by the sequence $\{(\epsilon, 1)\}_{\epsilon \downarrow 0}$ whose limit is the infeasible pair $(0, 1)$.

What is missing?

Supplementary visualization: non-attainment and the non-closed feasible set



On the feasible region the objective is the paraboloid $x_1^2 + 2(1 - x_2)^2 + 1.5$, whose valley floor is the line $x_1 = 0$.

That floor is the *infeasible slit* $\{x_1 = 0,\ x_2 \neq 0\}$ — cut out of the surface.

The sequence $(\epsilon, 1)$ slides down to the rim value 1.5, but its limit $(0, 1)$ falls in the slit. The infimum 1.5 is *not attained*: the feasible set is not closed.

Upper semicontinuity. Let each $\phi_{ij} : \mathbb{R}^n \to \mathbb{R}$ and $\psi_{ij} : \mathbb{R}^n \to \mathbb{R}$ be continuous functions for $j = 1, \ldots, J_i$ and $i = 0, \ldots, I$. Given a point $\bar{\mathbf{x}} \in \mathbb{R}^n$, if $\exists$ a neighborhood $\mathcal{N}$ of $\bar{\mathbf{x}}$ such that for every

$$(i, j) \in \mathcal{J}_0 \triangleq \big\{(i, j) \mid \phi_{ij}(\bar{\mathbf{x}}) = 0 \geq \psi_{ij}(\bar{\mathbf{x}}),\ i \in [I],\ j \in [J_i]\big\},$$

either of the following two conditions hold:

(A) for all $\mathbf{x} \in \mathcal{N}$, $\phi_{ij}(\mathbf{x})$ is nonnegative if $\psi_{ij}(\bar{\mathbf{x}}) < 0$,

(B) $\psi_{ij}(\mathbf{x})$ is nonnegative for all $\mathbf{x} \in \mathcal{N}$;

then the functions

$$\mathbf{x} \mapsto \sum_{j=1}^{J_i} \psi_{ij}(\mathbf{x}) \mathbb{1}_{[0,\infty)}\big(\phi_{ij}(\mathbf{x})\big)$$

are upper semicontinuous at $\bar{\mathbf{x}}$ for all $i = 0, 1, \ldots, I$.

Existence of a global maximizer

Suppose that c is continuous and its negative is coercive on the nonempty set X_{GHS} (which is not necessarily closed). Let $\{\mathbf{x}^\nu\}$ be a feasible sequence (which must exist) such that

$$\lim_{\nu \to \infty} \theta_{\mathrm{GHS}}(\mathbf{x}^\nu) = \sup_{\mathbf{x} \in X_{\mathrm{GHS}}} \theta_{\mathrm{GHS}}(\mathbf{x}) \in (-\infty, \infty). \tag{4}$$

Any accumulation point $\mathbf{x}^\infty$ of $\{\mathbf{x}^\nu\}$ (at least one of which must exist) satisfying condition (A) or (B) in Lemma must be a global maximizer of θ_{GHS} on X_{GHS}.

Some remarks

- Stationary conditions are necessary for a local minimizer; interested in sharp stationary solutions.
- Stationary solutions are more realistic computational targets for nonconvex problems.
- Stationarity sufficiency: sufficient conditions for a stationary solution to be locally minimizing.

Two stationarity concepts

- Pseudo Bouligand stationarity: based on a pull-out idea leading to a fixed-point definition.
- Epi-stationarity: based on an epigraphical formulation by lifting the problem.

Given a feasible $\bar{\mathbf{x}} \in X$, define 4 index sets

$$\mathcal{J}_{i>}(\bar{\mathbf{x}}) \triangleq \{j \in [J_i] \mid \phi_{ij}(\bar{\mathbf{x}}) > 0\}; \quad \mathcal{J}_{i=}(\bar{\mathbf{x}}) \triangleq \{j \in [J_i] \mid \phi_{ij}(\bar{\mathbf{x}}) = 0\}$$

$$\mathcal{J}_{i<}(\bar{\mathbf{x}}) \triangleq \{j \in [J_i] \mid \phi_{ij}(\bar{\mathbf{x}}) < 0\}; \quad \mathcal{J}_{i\geq}(\bar{\mathbf{x}}) \triangleq \{j \in [J_i] \mid \phi_{ij}(\bar{\mathbf{x}}) \geq 0\}$$

and the pulled-out problem (a standard NLP):

$$\underset{\mathbf{x} \in P}{\textbf{maximize}} \quad \theta(\mathbf{x}; \bar{\mathbf{x}}) \triangleq c(\mathbf{x}) + \sum_{j \in \mathcal{J}_{0\geq}(\bar{\mathbf{x}})} \psi_{0j}(\mathbf{x})$$

$$\textbf{subject to} \quad \left\{ \begin{array}{l} \text{for all } i = 1, \ldots, I \\ \mathbf{A}_{i\bullet}\mathbf{x} + \displaystyle\sum_{j \in \mathcal{J}_{i\geq}(\bar{\mathbf{x}})} \psi_{ij}(\mathbf{x}) \geq \eta_i \end{array} \right.$$

$$\textbf{and} \quad \left\{ \begin{array}{l} \text{for all } i = 0, 1, \ldots, I: \\ \phi_{ij}(\mathbf{x}) \geq 0 \quad \forall\, j \in \mathcal{J}_{i\geq}(\bar{\mathbf{x}}) \\ \phi_{ij}(\mathbf{x}) \leq 0 \quad \forall\, j \in \mathcal{J}_{i<}(\bar{\mathbf{x}}) \end{array} \right.$$

Definition: $\bar{\mathbf{x}}$ is a pseudo Bouligand stationary solution of the G-HSCOP if $\bar{\mathbf{x}}$ is a Bouligand stationary solution of the above NLP.

Recall

$$\bar{\mathbf{x}} \text{ is epi-stationary of } \underset{\mathbf{x}\in X}{\textbf{minimize}}\, f(\mathbf{x}) \quad \overset{\text{def}}{\Longleftrightarrow}$$

$$(\bar{t}, \bar{\mathbf{x}}) \text{ is Bouligand stationary of } \underset{(t,\mathbf{x})\in\mathbb{R}\times X}{\textbf{minimize}}\, t \quad \textbf{subject to} \quad t \geq f(\mathbf{x}).$$

It holds that

$$\text{local min} \;\Rightarrow\; \underbrace{\text{epi-stationary}}_{\text{theoretically sharp}} \;\Rightarrow\; \underbrace{\text{pseudo Bouligand stationary}}_{\text{provably computable}}.$$

Conversely, if the set $Z \triangleq \text{epi}(x) \cap (\mathbb{R} \times X)$[1] is convex-like near the pair $(f(\bar{\mathbf{x}}), \bar{\mathbf{x}}) \in Z$, then $\bar{\mathbf{x}}$ is a local minimizer of f on X if and only if $\bar{\mathbf{x}}$ is an epi-stationary solution.

[1] As shown on the slide; likely a typo for $\text{epi}(f) = \{(t, \mathbf{x}) : t \geq f(\mathbf{x})\}$, the feasible set of the lifted problem above.

For simplification, consider the G-HSCOP with a single Heaviside constraint:

$$\underset{\mathbf{x} \in P}{\textbf{minimize}} \quad \Phi(\mathbf{x}) \triangleq c(\mathbf{x}) + \sum_{k=1}^{K} \varphi_k(\mathbf{x}) \mathbb{1}_{[0,\infty)}\big(g_k(\mathbf{x})\big)$$

$$\textbf{subject to} \quad \sum_{\ell=1}^{L} \varphi_\ell(\mathbf{x}) \mathbb{1}_{[0,\infty)}\big(h_\ell(\mathbf{x})\big) \leq b \qquad \text{the functional constraint}$$

The associated "pulled-out" problem:

$$\begin{array}{rl} \textbf{minimize}_{\mathbf{x}} & \Phi(\mathbf{x};\bar{\mathbf{x}}) \triangleq c(\mathbf{x}) + \displaystyle\sum_{k\in\mathcal{K}_>(\bar{\mathbf{x}})} \varphi_k(\mathbf{x}) \\ \textbf{subject to} & \mathbf{x} \in P; \quad \displaystyle\sum_{\ell\in\mathcal{L}_>(\bar{\mathbf{x}})} \varphi_\ell(\mathbf{x}) \leq b \\ \textbf{and} & \left\{ \begin{array}{ll} g_k(\mathbf{x}) \leq 0 & \forall k \in \mathcal{K}_=(\bar{\mathbf{x}}) \cup \mathcal{K}_<(\bar{\mathbf{x}}) \triangleq \mathcal{K}_\leq(\bar{\mathbf{x}}) \\ g_k(\mathbf{x}) \geq 0 & \forall k \in \mathcal{K}_>(\bar{\mathbf{x}}) \\ h_\ell(\mathbf{x}) \leq 0 & \forall \ell \in \mathcal{L}_=(\bar{\mathbf{x}}) \cup \mathcal{L}_<(\bar{\mathbf{x}}) \triangleq \mathcal{L}_\leq(\bar{\mathbf{x}}) \\ h_\ell(\mathbf{x}) \geq 0 & \forall \ell \in \mathcal{L}_>(\bar{\mathbf{x}}) \end{array} \right. \end{array}$$

whose feasible set, excluding the functional constraint, is denoted $\widehat{\mathcal{S}}_{\rm ps}(\bar{\mathbf{x}})$.

$$\underset{\mathbf{x}\in\mathbb{R}^n}{\textbf{minimize}}\, f(\mathbf{x}) \qquad f : \mathbb{R}^n \to \mathbb{R} \text{ (scalar-valued)}$$

$$\Big\downarrow \text{ loc. min, } f \text{ diff.}$$

$$\underbrace{\nabla f(\bar{\mathbf{x}}) = \mathbf{0}}_{\text{equation}} \quad \Longleftrightarrow \quad \bar{\mathbf{x}} = \bar{\mathbf{x}} - \nabla f(\bar{\mathbf{x}})$$

$$\Big\downarrow$$

$$\boxed{\mathbf{x} = F(\mathbf{x})}, \quad F(\mathbf{x}) = \mathbf{x} - \nabla f(\mathbf{x}) \qquad F : \mathbb{R}^n \to \mathbb{R}^n \text{ (vector-valued)}$$

$$\mathbf{x} = \mathrm{Proj}_{\mathcal{X}}\big(\mathbf{x} - \nabla f(\mathbf{x})\big) \qquad \mathbf{x} \in \mathcal{X}$$

The penalized epigraphical formulation:

Assume that the functions φ_k and φ_ℓ are nonnegative on $P \cap g_k^{-1}(0)$ and $P \cap h_\ell^{-1}(0)$, respectively.

$$\underset{\mathbf{x}\in P;\, \mathbf{t},\mathbf{s}}{\textbf{minimize}} \quad \Phi_\lambda(\mathbf{x},\mathbf{t},\mathbf{s}) \triangleq \underbrace{c(\mathbf{x}) + \sum_{k=1}^{K} t_k}_{\Phi(\mathbf{x}) \text{ in epi-form}} + \lambda \underbrace{\max\left(\sum_{\ell=1}^{L} s_\ell - b,\ 0\right)}_{\substack{\text{constraint residual} \\ \text{fnc. in epi-form}}}$$

$$\textbf{subject to} \quad \begin{cases} \text{for all } k \in [K]: \\ \min\big(\max(\varphi_k(\mathbf{x}) - t_k,\ -g_k(\mathbf{x})),\ \max(g_k(\mathbf{x}),\ -t_k)\big) \le 0 \end{cases}$$

$$\begin{cases} \text{for all } \ell \in [L]: \\ \min\big(\max(\varphi_\ell(\mathbf{x}) - s_\ell,\ -h_\ell(\mathbf{x})),\ \max(h_\ell(\mathbf{x}),\ -s_\ell)\big) \le 0 \end{cases}$$

Exact penalization for finite λ

Theorem. Let c and φ_k be Lipschitz continuous. If $\{\bar{\mathbf{x}}, \bar{\mathbf{t}}, \bar{\mathbf{s}}\}$ is a B-stationary tuplet corresponding to a λ satisfying

$$\lambda > \mathrm{Lip}_c + K\,\mathrm{Lip}_\varphi,$$

then $\bar{\mathbf{x}}$ is a pseudo B-stationary solution of the Heaviside problem, provided that $\exists$ a vector $\mathbf{v} \in \mathcal{T}\big(\widehat{\mathcal{S}}_{\mathrm{ps}}(\bar{\mathbf{x}}); \bar{\mathbf{x}}\big)$ with unit length satisfying:

$$\sum_{\ell \in \mathcal{L}_>(\bar{\mathbf{x}})} \varphi_\ell'(\bar{\mathbf{x}}; \mathbf{v}) \leq -1 \quad \left[\text{a descent condition for } \textstyle\sum_{\ell \in \mathcal{L}_>(\bar{\mathbf{x}})} \varphi_\ell(\bullet) \text{ at } \bar{\mathbf{x}}\right].$$

For computation, apply smoothing of the open Heaviside function $\mathbb{1}_{(0,\infty)}(\bullet)$, in addition to penalization. There are 2 general approximation approaches that are “equivalent”:

(i) truncation based, and

(ii) average (i.e., integration) based.

Further, let (a difference-of-convex PA function)

$$T_{[0,1]}(t) \triangleq \min\{\max(t,0),\, 1\} = \max\{\min(t,1),\, 0\}$$
$$= \max(t,0) - \max(t-1,0), \quad t \in \mathbb{R}$$

be the truncation operator to the range $[0,1]$ and define the composite function:

$$\theta_{\mathrm{tr}}(t,\delta) \triangleq T_{[0,1]}\big(\widehat{\theta}(t,\delta)\big), \qquad (t,\delta) \in \mathbb{R} \times \mathbb{R}_{++}.$$

Example: The symmetric truncation hinge loss function for $\delta > 0$:

$$T_h(t,\delta) \triangleq \frac{1}{2\delta}\big[\max(t+\delta,0) - \max(t-\delta,0)\big]$$

fails condition (T2) and as a result does not "recover" the Heaviside function as $\delta \downarrow 0$ because $T_h(0,\delta) = \frac{1}{2}$ for all δ. Instead, the asymmetric modification remedies this failure:

$$\widetilde{T}_h(t,\delta) \triangleq \min\left\{\max\left(\frac{t}{\delta+\sqrt{\delta}} + \frac{\sqrt{\delta}}{1+\sqrt{\delta}},\, 0\right),\, 1\right\}.$$

Supplementary note. The truncation operator $T_{[0,1]}$ is difference-of-convex; so are its equivalent $\min/\max$ forms.

Consider the approximation of the Heaviside composite program:

$$\underset{\substack{\boxed{\mathbf{x} \in P} \\ \text{simple}}}{\textbf{minimize}} \quad \widehat{\Phi}_\lambda(\mathbf{x}, \delta) \triangleq c(\mathbf{x}) + \underbrace{\sum_{k=1}^{K} \varphi_k(\mathbf{x}) \theta_k^{\varphi}(g_k(\mathbf{x}), \delta)}_{\text{approx. of Heaviside in objective}}$$

$$+ \underset{\text{fixed}}{\lambda} \; \overbrace{\max\Big(\underbrace{\sum_{\ell=1}^{L} \varphi_\ell(\mathbf{x}) \theta_\ell^{\varphi}(h_\ell(\mathbf{x}), \delta)}_{\text{approx. of Heaviside in constraint}} - b, \; 0 \Big)}^{\text{constraint penalization}}$$

This is a difference-of-convex program, if all the functions are dc.

Goal: establish the limit property of a convergent sequence $\{\mathbf{x}^\nu\}$ of directional stationary solutions corresponding to a sequence $\{\delta_\nu\} \downarrow 0$; where

$$\widehat{\Phi}_\lambda(\bullet, \delta)'(\mathbf{x}^\nu; \mathbf{x} - \mathbf{x}^\nu) \geq 0, \qquad \text{for all } \mathbf{x} \in P.$$

(C4) descent for exact penalization: For any $\mathbf{x} \in P \cap \mathcal{N}$, where $\mathcal{N}_*$ such that $\sum_{\ell \in \mathcal{L}_>(\mathbf{x})} \varphi_\ell(\mathbf{x}) > b$, there exists a vector $\bar{\mathbf{v}} \in \mathcal{T}\big(\widehat{\mathcal{S}}_{\text{ps}}(\mathbf{x}); \mathbf{x}\big)$ with unit length satisfying:

$$\sum_{\ell \in \mathcal{L}_>(\mathbf{x})} \varphi'_\ell(\mathbf{x}; \bar{\mathbf{v}}) \leq -1; \quad \text{and}$$

(C5) Clarke regularity: the functions c, $\{\varphi_k\}_{k \in \mathcal{K}_>(\mathbf{x}^*)}$, and $\{\varphi_\ell\}_{\ell \in \mathcal{L}_>(\mathbf{x}^*)}$ are Clarke regular at $\mathbf{x}^*$. For a B-differentiable function f, this means that for all sequences $\{\mathbf{z}^\nu\}$ converging to $\bar{\mathbf{x}}$, it holds that

$$\limsup_{\nu \to \infty} f'(\mathbf{z}^\nu; \mathbf{v}) \leq f'(\bar{\mathbf{x}}; \mathbf{v}), \qquad \forall\, \mathbf{v} \in \mathbb{R}^n.$$

Theorem. Under (C1)–(C5), $\mathbf{x}^*$ is a pseudo B-stationary solution of the Heaviside optimization problem.

Chapter V

The A-HSCOP with mixed signed constant coefficients $\{\psi_{ik}\}$:

$$\begin{array}{ll} \underset{\mathbf{x}\in P}{\textbf{maximize}} & \Phi(\mathbf{x}) \triangleq c(\mathbf{x}) + \sum_{k=1}^{K_0} \psi_{0k}\, \mathbb{1}_{[0,\infty)}\big(\phi_{0k}(\mathbf{x})\big) \\ \textbf{subject to} & \text{for all } i = 1, \ldots, I \\ & \mathbf{A}_{i\bullet}\mathbf{x} + \sum_{k=1}^{K_i} \psi_{ik}\, \mathbb{1}_{[0,\infty)}\big(\phi_{ik}(\mathbf{x})\big) \geq \eta_i. \end{array}$$

Feasible set denoted X_{AHS}.

Proposition: If P is convex and each ϕ_{ij} is piecewise affine, then X_{AHS} is locally star-shaped; i.e., for every $\bar{\mathbf{x}} \in X_{\mathrm{AHS}}$, $\exists$ a neighborhood of $\bar{\mathbf{x}}$ such that for every $\mathbf{x} \in X_{\mathrm{AHS}}$, a scalar $\bar{\tau} > 0$ exists satisfying $\bar{\mathbf{x}} + \tau(\mathbf{x} - \bar{\mathbf{x}}) \in X_{\mathrm{AHS}}$ for all $\tau \in [0, \bar{\tau}]$.

This is a sharper property than local convexity-like.

Suppose that c is B(ouligand) differentiable and locally concave-like near $\bar{\mathbf{x}} \in X^{\mathrm{PA}}_{\mathrm{AHS}}$, i.e., if

$$f(\mathbf{x}) \leq f(\bar{\mathbf{x}}) + f'(\bar{\mathbf{x}}; \mathbf{x} - \bar{\mathbf{x}}) \quad \forall\, \mathbf{x} \text{ sufficiently close to } \bar{\mathbf{x}}.$$

Then (A) $\Leftrightarrow$ (B) $\Leftrightarrow$ (C), where

(A) $\bar{\mathbf{x}}$ is a local maximizer of the A-HSCOP;

(B) $\bar{\mathbf{x}}$ is an epi-stationary point of the A-HSCOP;

(C) for all $\mathbf{x} \in X^{\mathrm{PA}}_{\mathrm{AHS}}$ sufficiently near $\bar{\mathbf{x}}$, it holds that

$$\sum_{j=1}^{J_0} \psi_{0j}\, \mathbb{1}_{[0,\infty)}\big(\phi_{0j}(\mathbf{x})\big) \leq \sum_{j=1}^{J_0} \psi_{0j}\, \mathbb{1}_{[0,\infty)}\big(\phi_{0j}(\bar{\mathbf{x}})\big);$$

moreover, if equality holds in any one of the above 3 inequalities, then $c(\mathbf{x}) \leq c(\bar{\mathbf{x}})$ and $c'(\bar{\mathbf{x}}; \mathbf{x} - \bar{\mathbf{x}}) \leq 0$.

A hierarchical interpretation: $\bar{\mathbf{x}}$ locally maximizes the Heaviside sum, followed by the function c among the $\mathbf{x}$ with maximum Heaviside sum.

A restrictive but motivating existence result

Let P be a closed set. Suppose that c is continuous and negatively coercive on the nonempty set X_{AHS} and that the functions $\{\phi_{ij}\}$ are continuous. If

$$\boxed{\exists\, \bar{\mathbf{x}} \in X_{\text{AHS}} \text{ satisfying } \phi_{ij}(\bar{\mathbf{x}}) < 0 \text{ for all } (i,j) \text{ such that } \psi_{ij} < 0,}$$

then there exists $\bar{\varepsilon} > 0$ such that the problem:

$$\underset{\mathbf{x} \in P}{\textbf{maximize}} \quad c(\mathbf{x}) + \sum_{j=1}^{J_0} \psi_{0j}^{+} \mathbb{1}_{[0,\infty)}\big(\phi_{0j}(\mathbf{x})\big) - \sum_{j=1}^{J_0} \psi_{0j}^{-} \mathbb{1}_{(-\varepsilon,\infty)}\big(\phi_{0j}(\mathbf{x})\big)$$

$$\textbf{subject to} \quad \text{for all } i = 1, \ldots, I:$$

$$\mathbf{A}_{i\bullet}\mathbf{x} + \sum_{j=1}^{J_i} \psi_{ij}^{+} \mathbb{1}_{[0,\infty)}\big(\phi_{ij}(\mathbf{x})\big) - \sum_{j=1}^{J_i} \psi_{ij}^{-} \mathbb{1}_{(-\varepsilon,\infty)}\big(\phi_{ij}(\mathbf{x})\big) \geq \eta_i,$$

where $\psi_{ij}^{\pm} \triangleq \max(0, \pm\psi_{ij})$, has an optimal solution for all $\varepsilon \in (0, \bar{\varepsilon}]$.

The local semicontinuous equivalent $(\mathrm{P}^{\varepsilon}_{\mathrm{AHS}})$

For a given $\varepsilon > 0$,

$$\operatorname*{\textbf{maximize}}_{\mathbf{x}\in P} \quad \theta^{\varepsilon}_{\mathrm{AHS}} \triangleq c(\mathbf{x}) + \underbrace{\sum_{j=1}^{J_0} \psi^{+}_{0j} \mathbb{1}_{[0,\infty)}\big(\phi_{0j}(\mathbf{x})\big) - \sum_{j=1}^{J_0} \psi^{-}_{0j} \mathbb{1}_{(-\varepsilon,\infty)}\big(\phi_{0j}(\mathbf{x})\big)}_{\text{upper semicontinuous}}$$

subject to for all $i = 1, \ldots, I$:

$$\mathbf{A}_{i\bullet}\mathbf{x} + \underbrace{\sum_{j=1}^{J_i} \psi^{+}_{ij} \mathbb{1}_{[0,\infty)}\big(\phi_{ij}(\mathbf{x})\big) - \sum_{j=1}^{J_i} \psi^{-}_{ij} \mathbb{1}_{(-\varepsilon,\infty)}\big(\phi_{ij}(\mathbf{x})\big)}_{\text{upper semicontinuous}} \geq \eta_i,$$

feasible set denoted $X^{\varepsilon}_{\mathrm{AHS}}$, including P:

$$X^{\varepsilon'}_{\mathrm{AHS}} \subseteq X^{\varepsilon}_{\mathrm{AHS}} \subseteq X_{\mathrm{AHS}}, \qquad \forall\, \varepsilon' > \varepsilon > 0 \qquad \text{[approximating from inside].}$$

With local sign-invariance assumption.

Let $\bar{\mathbf{x}} \in X$ given. Let each ϕ_{ij} be continuous. There exists $\bar{\varepsilon} > 0$ such that for all $\varepsilon \in (0, \bar{\varepsilon}]$, the following two statements hold.

(A) If $\bar{\mathbf{x}}$ is a local maximizer of the A-HSCOP, then $\bar{\mathbf{x}}$ is a local maximizer of $(\mathrm{P}^{\varepsilon}_{\mathrm{AHS}})$.

(B) Conversely, suppose that $\bar{\mathbf{x}}$ satisfies the local sign invariance property with reference to $\big\{(\psi_{ij}, \phi_{ij})_{j=1}^{J_i}\big\}_{i=0}^{I}$; i.e., ϕ_{ij} is nonnegative near $\bar{\mathbf{x}}$ for all $j \in \mathcal{J}^{-}_{i,0}(\bar{\mathbf{x}})$ and all $i = 0, 1, \ldots, I$. If $\bar{\mathbf{x}}$ is a local maximizer of $(\mathrm{P}^{\varepsilon}_{\mathrm{AHS}})$, then $\bar{\mathbf{x}}$ is a local maximizer of the A-HSCOP.

Chapter VI

IP-based Solution Algorithms for scalar combinations

sketch of key ideas; details to be provided in implementation

Classes of Problems:

- Scalar combinations: constant ψ_{ij}
  - $\psi_{ij} \geq 0$ and concave inner functions ϕ_{ij}
  - $\psi_{ij} \geq 0$ and piecewise affine inner functions ϕ_{ij}
  - mixed-signed ψ_{ij} and PA ϕ_{ij}
  - mixed-signed ψ_{ij} and multiplicative HSCOP with concave or PA ϕ_{ij}

Blanket assumption: $\exists\, M > 0$ such that

$$|\phi_{ij}(\mathbf{x})| \leq M \quad \forall\,(i,j) \text{ and all } \mathbf{x} \in P.$$

Key restriction: Avoid solving non-concave or non-PA integer minimization problems, because the technology is not here yet.

Base case: $\psi_{ij} \geq 0$ and concave ϕ_{ij}. The full IP formulation:

$$\underset{\mathbf{x} \in P;\, \mathbf{z}}{\textbf{maximize}} \quad \theta^{\oplus}_{\mathrm{AHS}}(\mathbf{x}, \mathbf{z}) \triangleq c(\mathbf{x}) + \sum_{j=1}^{J_0} \psi_{0j} z_{0j}$$

$$\textbf{subject to} \quad \text{for all } i = 0, 1, \ldots, I :$$

$$\mathbf{A}_{i\bullet}\mathbf{x} + \sum_{j=1}^{J_i} \psi_{ij} z_{ij} \geq \eta_i$$

$$\textbf{and} \quad \left\{ \begin{array}{l} \underbrace{\phi_{ij}(\mathbf{x}) \geq -M(1 - z_{ij})}_{\text{convex constraints}} \\ z_{ij} \in \{0, 1\} \end{array} \right\} \; j = 1, \ldots, J_i$$

Equivalence: $\bar{\mathbf{x}}$ is optimal for the A-HSCOP if and only if $(\bar{\mathbf{x}}, \bar{\mathbf{z}})$, where $\bar{z}_{ij} \triangleq \mathbb{1}_{[0,\infty)}(\phi_{ij}(\bar{\mathbf{x}}))$, is optimal for IP.

The PIP subproblems. Goal is to reduce the number of integer variables.

For a given scalar $\delta > 0$ and vector $\bar{\mathbf{x}} \in X_{\mathrm{AHS}}$, let $\mathcal{J}^{\delta}_{i;>}(\bar{\mathbf{x}})$ and $\mathcal{J}^{\delta}_{i;<}(\bar{\mathbf{x}})$ be two disjoint subsets of $\{1, \ldots, J_i\}$ for all $i = 0, 1, \ldots, I$ such that

$$\begin{aligned} \mathcal{J}^{\delta}_{i;>}(\bar{\mathbf{x}}) &\subseteq \underbrace{\left\{ j \in [J_i] \mid \phi_{ij}(\bar{\mathbf{x}}) \geq \delta \right\}}_{\text{expansion of } \phi_{ij}(\bar{\mathbf{x}}) \geq 0} \\ \mathcal{J}^{\delta}_{i;<}(\bar{\mathbf{x}}) &\subseteq \underbrace{\left\{ j \in [J_i] \mid \phi_{ij}(\bar{\mathbf{x}}) \leq -\delta \right\}}_{\text{expansion of } \phi_{ij}(\bar{\mathbf{x}}) \leq 0} \end{aligned} \tag{5}$$

and let $\mathcal{J}^{\delta}_{i;0}(\bar{\mathbf{x}})$ be the complement of $\mathcal{J}^{\delta}_{i;>}(\bar{\mathbf{x}}) \cup \mathcal{J}^{\delta}_{i;<}(\bar{\mathbf{x}})$; thus

$$\mathcal{J}^{\delta}_{i;0}(\bar{\mathbf{x}}) \supseteq \underbrace{\left\{ j \in [J_i] \mid -\delta < \phi_{ij}(\bar{\mathbf{x}}) < \delta \right\}}_{\text{indeterminate indices}} \supseteq \left\{ j \in [J_i] \mid \phi_{ij}(\bar{\mathbf{x}}) = 0 \right\}.$$

$\mathcal{J}^{\delta}_{i;>}(\bar{\mathbf{x}})$, $\mathcal{J}^{\delta}_{i;<}(\bar{\mathbf{x}})$, and $\mathcal{J}^{\delta}_{i;0}(\bar{\mathbf{x}})$ are working index sets that determine the binary variables to be fixed or solved.

For a given scalar $\rho > 0$, define the $\left(\mathrm{IP}^{\oplus;\delta}_{\mathrm{AHS};\rho}(\bar{\mathbf{x}})\right)$:

$$\underset{\mathbf{x}\in P;\,\mathbf{z}}{\textbf{maximize}} \quad \theta^{\oplus;\delta}_{\mathrm{AHS};\rho}(\mathbf{x},\mathbf{z};\bar{\mathbf{x}}) \triangleq c(\mathbf{x}) + \sum_{j\in\mathcal{J}^{\delta}_{0;0}(\bar{\mathbf{x}})} \psi_{0j} z_{0j} - \underbrace{\frac{\rho}{2}\|\mathbf{x}-\bar{\mathbf{x}}\|_2^2}_{\text{proximal term}}$$

subject to for all $i = 1,\ldots,I$:

$$\mathbf{A}_{i\bullet}\mathbf{x} + \sum_{j\in\mathcal{J}^{\delta}_{i;0}(\bar{\mathbf{x}})} \psi_{ij} z_{ij} + \sum_{j\in\mathcal{J}^{\delta}_{i;>}(\bar{\mathbf{x}})} \psi_{ij} \geq \eta_i,$$

and for all $i = 0, 1, \ldots, I$:

$$\left.\begin{array}{l} \phi_{ij}(\mathbf{x}) \geq -M(1-z_{ij}) \\ z_{ij} \in \{0,1\} \end{array}\right\} \; j \in \mathcal{J}^{\delta}_{i;0}(\bar{\mathbf{x}}) \quad \text{integer variables here only}$$

$$\phi_{ij}(\mathbf{x}) \geq 0 \text{ (equivalent to fixing } z_{ij} = 1); \;\; j \in \mathcal{J}^{\delta}_{i;>}(\bar{\mathbf{x}})$$

$$\phi_{ij}(\mathbf{x}) \text{ free (equivalent to fixing } z_{ij} = 0); \;\; j \in \mathcal{J}^{\delta}_{i;<}(\bar{\mathbf{x}})$$

This is a convex-constrained mixed-binary program, solvable by GUROBI.

The PIP method for the A-HSCOP with $\psi_{ij} \geq 0$ and concave ϕ_{ij}

Main computation. Given an iterate $\mathbf{x}^\nu$, select two disjoint index sets $\mathcal{J}^{\delta}_{i;>}(\mathbf{x}^\nu)$ and $\mathcal{J}^{\delta}_{i;<}(\mathbf{x}^\nu)$ satisfying the inclusions in (5). Solve the partial $\left(\mathrm{IP}^{\oplus;\delta}_{\mathrm{AHS};\rho}(\mathbf{x}^\nu)\right)$, warm starting at $\mathbf{x}^\nu$.

Termination. If $\mathbf{x}^\nu$ itself is an optimal solution of the self-defined partial IP, stop; the vector $\mathbf{x}^\nu$ is a local maximizer of the A-HSCOP.

Updates of index sets. Otherwise, let $\mathbf{x}^{\nu+1}$ be an optimal solution and choose new index sets $\mathcal{J}^{\delta}_{i;>}(\mathbf{x}^{\nu+1})$ and $\mathcal{J}^{\delta}_{i;<}(\mathbf{x}^{\nu+1})$ satisfying (5), ensuring that the cardinality

$$\left| \mathcal{J}^{\delta}_{i;>}(\mathbf{x}^{\nu+1}) \cup \mathcal{J}^{\delta}_{i;<}(\mathbf{x}^{\nu+1}) \right|$$

of the new sets is at least one less than the cardinality of the old sets at $\mathbf{x}^\nu$. Let $\nu \leftarrow \nu + 1$ and repeat the main computational step.

Note: (1) The index sets ensure that the iterate $\mathbf{x}^{\nu+1}$ will remain feasible to its self-defined IP; and (2) expansions of the index sets are subject to variations in implementation.

The case of the A-HSCOP with $\psi_{ij} \geq 0$ and PA ϕ_{ij}

$$\phi_{ij}^{\mathrm{PA}}(\mathbf{x}) = \underbrace{\max_{1 \leq k \leq K_{ij}} \left[(\mathbf{a}_{ij}^k)^\top \mathbf{x} + \alpha_{ij}^k \right]}_{\text{denoted } \phi_{ij}^{\mathrm{PAcvx}}(\mathbf{x})} + \underbrace{\min_{1 \leq \ell \leq L_{ij}} \left[(\mathbf{b}_{ij}^\ell)^\top \mathbf{x} + \beta_{ij}^\ell \right]}_{\text{denoted } \phi_{ij}^{\mathrm{PAcve}}(\mathbf{x})}$$

$$= \max_{1 \leq k \leq K_{ij}} \underbrace{\left[(\mathbf{a}_{ij}^k)^\top \mathbf{x} + \alpha_{ij}^k + \phi_{ij}^{\mathrm{PAcve}}(\mathbf{x}) \right]}_{\text{denoted } \phi_{ij}^{k;+}(\mathbf{x}), \text{ each concave}}$$

$$\geq \phi_{ij}^{k;+}(\mathbf{x}) \quad \forall k \in [K_{ij}],$$

for some constant vectors $\mathbf{a}_{ij}^k$ and $\mathbf{b}_{ij}^\ell$ and scalars α_{ij}^k and β_{ij}^ℓ. For a given vector $\bar{\mathbf{x}}$, let for all pairs (i, j),

$$\mathcal{K}_{ij}(\bar{\mathbf{x}}) \triangleq \left\{ k \in [K_{ij}] \mid (\mathbf{a}_{ij}^k)^\top \bar{\mathbf{x}} + \alpha_{ij}^k \geq \phi_{ij}^{\mathrm{PAcvx}}(\bar{\mathbf{x}}) \right\}.$$

Note that $\phi_{ij}^{\mathrm{PA}}(\bar{\mathbf{x}}) = \phi_{ij}^{k;+}(\bar{\mathbf{x}})$ for any $k \in \widehat{\mathcal{K}}_{ij} \supseteq \mathcal{K}_{ij}(\bar{\mathbf{x}})$, in particular for $\widehat{\mathcal{K}}_{ij}$ given by

$$\widehat{\mathcal{K}}_{ij}^{\delta}(\bar{\mathbf{x}}) \triangleq \left\{ k \in [K_{ij}] \mid (\mathbf{a}_{ij}^k)^\top \bar{\mathbf{x}} + \alpha_{ij}^k \geq \phi_{ij}^{\mathrm{PAcvx}}(\bar{\mathbf{x}}) - \delta \right\}, \quad \delta > 0.$$

The regularized, decomposed A-HSCOP for a given tuple $\mathbf{k} \triangleq (k_{ij})$:

$$\begin{aligned}
\underset{\mathbf{x} \in P}{\textbf{maximize}} \quad & \theta^{\oplus;k_0}_{\mathrm{AHS};\rho}(\mathbf{x}) \triangleq c(\mathbf{x}) + \sum_{j=1}^{J_0} \psi_{0j} \mathbb{1}_{[0,\infty)}\big(\phi^{k_{0j};+}_{0j}(\mathbf{x})\big) \\
& \qquad - \frac{\rho}{2}\|\mathbf{x} - \bar{\mathbf{x}}\|_2^2; \quad \rho \geq 0 \\
\textbf{subject to} \quad & \text{for all } i = 1, \ldots, I: \\
& \mathbf{A}_{i\bullet}\mathbf{x} + \sum_{j=1}^{J_i} \psi_{ij} \mathbb{1}_{[0,\infty)}\big(\phi^{k_{ij};+}_{ij}(\mathbf{x})\big) \geq \eta_i.
\end{aligned}$$

feasible set denoted $X^{\oplus;\mathbf{k}}_{\mathrm{AHS}}$; problem denoted $\big(\mathrm{P}^{\oplus;\mathbf{k}}_{\mathrm{AHS};\rho}\big)$

Each is of the previous kind: nonnegative coefficients and concave inner functions; furthermore,

$$\theta_{\mathrm{AHS}}(\mathbf{x}) = \max_{\mathbf{k}} \theta^{\oplus;k_0}_{\mathrm{AHS};0}(\mathbf{x}), \quad \text{for any } \mathbf{x}; \quad \text{and} \quad X_{\mathrm{AHS}} = \bigcup_{\mathbf{k}} X^{\oplus;\mathbf{k}}_{\mathrm{AHS}}.$$

There are 2 versions of the algorithm:

- **Version I**: PIP first and then decompose each PIP subproblem
- **Version II**: Decompose first and then apply PIP to each decomposed problem.

Presented below is Version II: decompose-then-PIP.

The Iterative Decomposed Shrinkage Algorithm, IDSA-PIP, **for the A-HSCOP** with mixed-signed ψ_{ij} and PA ϕ_{ij}

Several steps:

- Write $\psi_{ij} = \psi_{ij}^{+} - \psi_{ij}^{-}$; note
$$-\psi_{ij}^{-}\mathbb{1}_{(-\varepsilon,\infty)}\big(\phi_{ij}(\mathbf{x})\big) = -\psi_{ij}^{-} + \psi_{ij}^{-}\mathbb{1}_{[0,\infty)}\big(-\phi_{ij}(\mathbf{x}) - \varepsilon\big)$$
- Solve a sequence of decomposed ε_ν-approximation problems by the PIP method described above, for a sequence of $\{\varepsilon_\nu\} \downarrow 0$.
- For each ε-subproblem, need majorization and minorization $\phi_{ij}^{\mathrm{PA}}(\mathbf{x})$ by
$$\phi_{ij}^{\ell;-}(\mathbf{x}) \le \phi_{ij}^{\mathrm{PA}}(\mathbf{x}) \le \phi_{ij}^{k;+}(\mathbf{x}),$$
respectively.
- After majorization/minorization, each PIP subproblem is a reduced mixed-binary, concave maximization program (with the only nonlinear function being $c(\mathbf{x})$).

The computational workhorse $\left(\mathrm{IP}^{\varepsilon_\nu;k,\ell,\delta'}_{\mathrm{AHS};\rho}(\mathbf{x}^\nu)\right)$ at $\widehat{\mathbf{x}} \in X^{\varepsilon}_{\mathrm{AHS}}$:

$$\begin{array}{rl}
\underset{\mathbf{x}\in P;\,\mathbf{z}^{\pm}}{\textbf{maximize}} & c(\mathbf{x}) + \displaystyle\sum_{j\in\widehat{\mathcal{J}}^{+\delta'}_{0;0}(\widehat{\mathbf{x}})} \psi^{+}_{0j} z^{+}_{0j} + \sum_{j\in\widehat{\mathcal{J}}^{-\delta'}_{0;0}(\widehat{\mathbf{x}})} \psi^{-}_{0j} z^{-}_{0j} + \sum_{j\in\widehat{\mathcal{J}}^{+\delta'}_{0;>}(\widehat{\mathbf{x}})} \psi^{+}_{0j} \\
& \quad - \displaystyle\sum_{j\in\widehat{\mathcal{J}}^{-\delta'}_{0;0}(\widehat{\mathbf{x}})\cup\widehat{\mathcal{J}}^{-\delta'}_{0;<}(\widehat{\mathbf{x}})} \psi^{-}_{0j} - \frac{\rho}{2}\|\mathbf{x}-\mathbf{x}^\nu\|_2^2 \\
\textbf{subject to} & \mathbf{A}_{i\bullet}\mathbf{x} + \displaystyle\sum_{j\in\widehat{\mathcal{J}}^{+\delta'}_{i;0}(\widehat{\mathbf{x}})} \psi^{+}_{ij} z^{+}_{ij} + \sum_{j\in\widehat{\mathcal{J}}^{+\delta'}_{i;>}(\widehat{\mathbf{x}})} \psi^{+}_{ij} + \sum_{j\in\widehat{\mathcal{J}}^{-\delta'}_{i;0}(\widehat{\mathbf{x}})} \psi^{-}_{ij} z^{-}_{ij} \\
& \quad \geq \displaystyle\sum_{j\in\widehat{\mathcal{J}}^{-\delta'}_{i;0}(\widehat{\mathbf{x}})\cup\widehat{\mathcal{J}}^{-\delta'}_{i;<}(\widehat{\mathbf{x}})} \psi^{-}_{ij} + \eta_i, \quad \forall i \in [I]:
\end{array}$$

and for all $i = 0, \ldots, I$:

$$\left.\begin{array}{l}
\phi^{k_{ij};+}_{ij}(\mathbf{x}) \geq -M(1-z^{+}_{ij});\ z^{+}_{ij}\in\{0,1\}; \qquad j \in \widehat{\mathcal{J}}^{+\delta'}_{i;0}(\widehat{\mathbf{x}}) \\
-\phi^{\ell_{ij};-}_{ij}(\mathbf{x}) \geq -M(1-z^{-}_{ij}) + \varepsilon;\ z^{-}_{ij}\in\{0,1\};\ j \in \widehat{\mathcal{J}}^{-\delta'}_{i;0}(\widehat{\mathbf{x}})
\end{array}\right\}$$

$$\left.\begin{array}{l}
\phi^{k_{ij};+}_{ij}(\mathbf{x}) \geq 0;\ (\text{equivalent to } z^{+}_{ij} = 1); \quad j \in \widehat{\mathcal{J}}^{+\delta'}_{i;>}(\widehat{\mathbf{x}}) \\
-\phi^{\ell_{ij};-}_{ij}(\mathbf{x}) \geq \varepsilon;\ (\text{equivalent to } z^{-}_{ij} = 1);\ j \in \widehat{\mathcal{J}}^{-\delta'}_{i;>}(\widehat{\mathbf{x}})
\end{array}\right\}$$

Sketch of IDSA-PIP

Initialization. Let $\{\varepsilon_\nu\}_{\nu=0}^{\infty} \downarrow 0$. Let $\mathbf{x}^0 \in X_{\mathrm{AHS}}^{\varepsilon_0}$; let $(\rho, \delta, \delta') > 0$, and $\nu = 0$.

Main Computation. Apply PIP to solve the problem $\left(\mathrm{P}_{\mathrm{AHS};\rho}^{\varepsilon_\nu;k,\ell}(\mathbf{x}^\nu)\right)$ for all pairs $(\mathbf{k}, \boldsymbol{\ell})$ in $\mathcal{M}^\delta(\mathbf{x}^\nu)$ and obtain a local maximizer, denoted $\mathbf{x}^{k,\ell}(\mathbf{x}^\nu)$. Determine the maximum of

$$\theta_{\mathrm{AHS};\rho}^{\varepsilon_\nu;k;\ell}\left(\mathbf{x}^{k,\ell}(\mathbf{x}^\nu);\mathbf{x}^\nu\right)$$

among all pairs $(\mathbf{k}, \boldsymbol{\ell})$ in $\mathcal{M}^\delta(\mathbf{x}^\nu)$.

Termination. Terminate if $\mathbf{x}^{\nu+1}$ satisfies a prescribed rule (e.g., if $\|\mathbf{x}^{\nu+1} - \mathbf{x}^\nu\|$ is within a given tolerance). Otherwise, return to the main computation step with $\nu \leftarrow \nu + 1$.

Let P be compact, c be continuous and each ϕ_{ij} be PA. Every accumulation point $\mathbf{x}^\infty$ of a sequence $\{\mathbf{x}^\nu\}$ produced by the IDSA-PIP algorithm is feasible to the A-HSCOP (with mixed signed ψ_{ij}).

Moreover, if $\mathcal{J}_{i,0}^-(\mathbf{x}^\infty) = \emptyset$ for all $i = 0, 1, \ldots, I$, $\exists$ a pair $(\mathbf{k}^\infty, \boldsymbol{\ell}^\infty) \in \mathcal{M}(\mathbf{x}^\infty)$ such that $\mathbf{x}^\infty$ is a local maximizer of the problem:

$$\begin{aligned}
\underset{\mathbf{x}\in P}{\textbf{maximize}} \quad & c(\mathbf{x}) + \sum_{j=1}^{J_0} \psi_{0j}^+ \mathbb{1}_{[0,\infty)}\left(\phi_{0j}^{k_{0j}^\infty;+}(\mathbf{x})\right) \\
& - \sum_{j=1}^{J_0} \psi_{0j}^- \mathbb{1}_{[0,\infty)}\left(\phi_{0j}^{\ell_{0j}^\infty;-}(\mathbf{x})\right) \\
\textbf{subject to} \quad & \mathbf{A}_{i\bullet}\mathbf{x} + \sum_{j=1}^{J_i} \psi_{ij}^+ \mathbb{1}_{[0,\infty)}\left(\phi_{ij}^{k_{ij}^\infty;+}(\mathbf{x})\right) \\
& - \sum_{j=1}^{J_i} \psi_{ij}^- \mathbb{1}_{[0,\infty)}\left(\phi_{ij}^{\ell_{ij}^\infty;-}(\mathbf{x})\right) \geq \eta_i, \quad \text{for all } i \in [I].
\end{aligned} \tag{6}$$

If additionally $\mathcal{M}(\mathbf{x}^\infty)$ is a singleton, then $\mathbf{x}^\infty$ is a local maximizer of the A-HSCOP.

The multiplicative Heaviside composite problem M-HSCOP:

$$\underset{\mathbf{x}\in P}{\textbf{maximize}} \quad c(\mathbf{x}) + \sum_{j=1}^{J_0} \psi_{0j}\, \mathbb{1}_{[0,\infty)}\big(\phi_{0j}(\mathbf{x})\big)\, \mathbb{1}_{(0,\infty)}\big(\varphi_{0j}(\mathbf{x})\big)$$

$$\textbf{subject to} \quad \text{for all } i = 1, \ldots, I:$$

$$\mathbf{A}_{i\bullet}\mathbf{x} + \sum_{j=1}^{J_i} \psi_{ij}\, \mathbb{1}_{[0,\infty)}\big(\phi_{ij}(\mathbf{x})\big)\, \mathbb{1}_{(0,\infty)}\big(\varphi_{ij}(\mathbf{x})\big) \geq \eta_i.$$

Approximation of the product:

$$\psi_{ij}\, \underbrace{\mathbb{1}_{[0,\infty)}\big(\phi_{ij}(\mathbf{x})\big)}_{\text{closed Heaviside}}\, \underbrace{\mathbb{1}_{(0,\infty)}\big(\varphi_{ij}(\mathbf{x})\big)}_{\text{open Heaviside}} \approx$$

$$\psi_{ij}^{+}\, \mathbb{1}_{[0,\infty)}\big(\phi_{ij}(\mathbf{x})\big)\, \mathbb{1}_{[\varepsilon,\infty)}\big(\varphi_{ij}(\mathbf{x})\big)$$

$$- \psi_{ij}^{-}\, \mathbb{1}_{(-\varepsilon,\infty)}\big(\phi_{ij}(\mathbf{x})\big)\, \mathbb{1}_{(0,\infty)}\big(\varphi_{ij}(\mathbf{x})\big)$$

preserves monotonicity and ensures applicability and guaranteed convergence of the IDSA-PIP algorithm. (vs. an alternative approximation).

Classes of Problems:

- Affine outer combinations of HSCOPs
- A-HSCOP with complementarity constraints
- Fractional HSCOPs and 2-sum fractional extension
- Doubly Heaviside composite problems

HSCOP with affine ψ_{ij}

$$\underset{\mathbf{x}\in P}{\textbf{maximize}} \quad \theta_{\mathrm{QHS}}(\mathbf{x}) \triangleq c(\mathbf{x}) + \sum_{j=1}^{J_0} \left[(\mathbf{a}^{0j})^\top \mathbf{x} + \alpha_{0j} \right] \mathbb{1}_{[0,\infty)}\big(\phi_{0j}(\mathbf{x})\big)$$

subject to for $i = 1, \ldots, I$:

$$\mathbf{A}_{i\bullet}\mathbf{x} + \sum_{j=1}^{J_i} \left[(\mathbf{a}^{ij})^\top \mathbf{x} + \alpha_{ij} \right] \mathbb{1}_{[0,\infty)}\big(\phi_{ij}(\mathbf{x})\big) \geq \eta_i, \quad i = 1, \ldots, I.$$

Several steps:

Step 1: Write $(\mathbf{a}^{ij})^\top \mathbf{x} + \alpha_{ij} = \left[(\mathbf{a}^{ij})^\top \mathbf{x} + \alpha_{ij}\right]_+ - \left[(\mathbf{a}^{ij})^\top \mathbf{x} + \alpha_{ij}\right]_-$,

$$\underset{\mathbf{x}\in P}{\textbf{maximize}} \;\; c(\mathbf{x}) + \sum_{j=1}^{J_0} \left\{ \begin{array}{l} \left[(\mathbf{a}^{0j})^\top \mathbf{x} + \alpha_{0j}\right]_+ \mathbb{1}_{[0,\infty)}\left(\phi_{0j}(\mathbf{x})\right) \\ - \left[(\mathbf{a}^{0j})^\top \mathbf{x} + \alpha_{0j}\right]_- \mathbb{1}_{[0,\infty)}\left(\phi_{0j}(\mathbf{x})\right) \end{array} \right\}$$

and similarly for constraints.

Step 2: Open up the negative Heaviside terms:

$$\left[(\mathbf{a}^{0j})^\top \mathbf{x} + \alpha_{0j}\right]_- \mathbb{1}_{[0,\infty)}\left(\phi_{0j}(\mathbf{x})\right) \approx \left[(\mathbf{a}^{0j})^\top \mathbf{x} + \alpha_{0j}\right]_- \mathbb{1}_{(-\varepsilon,\infty)}\left(\phi_{0j}(\mathbf{x})\right).$$

Step 3: Linearize the products: $s = yz$, where $0 \le y - s \le \bar{y}(1-z)$ and $0 \le s \le \bar{y}z$, where $0 \le y \le \bar{y}$ and $z \in \{0,1\}$.

Step 4: Apply IDSA-PIP to the resulting mixed-IP formulation.

Case Studies: Policy Learning and Classification via Affine Heaviside Composite Optimization

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Based on the research works:

- Solving Constrained Affine Heaviside Composite Optimization Problems by a Progressive IP Approach. (2026) `https://arxiv.org/abs/2605.06020` *Joint work with Ke Zheng, Yurui Wang, Jong-Shi Pang.*
- Offline policy learning with weight clipping and Heaviside composite optimization. (2026) `https://arxiv.org/abs/2601.12117`. *Joint work with Jingren Liu, Hanzhang Qin, Mabel C. Chou, Jong-Shi Pang.*
- Classification and treatment Learning with constraints via Heaviside optimization: a Progressive MIP Method (2025). Computational Optimization and Applications. *Joint work with Yue Fang and Jong-Shi Pang.*

Affine Heaviside composite optimization problem (A-HSCOP)

$$\underset{\mathbf{x}\in P}{\textbf{maximize}} \quad c(\mathbf{x}) + \sum_{j=1}^{J_0} \psi_{0j}\, \mathbb{1}_{[0,\infty)}\big(\phi_{0j}(\mathbf{x})\big),$$

$$\textbf{subject to} \quad \sum_{j=1}^{J_i} \psi_{ij}\, \mathbb{1}_{[0,\infty)}\big(\phi_{ij}(\mathbf{x})\big) \geq b_i, \quad \text{for } i = 1, \ldots, m \quad \text{(upper semicontinuous)}$$

- Heaviside function $\mathbb{1}_{[0,\infty)}(t) \triangleq \begin{cases} 1 & \text{if } t \geq 0 \\ 0 & \text{otherwise} \end{cases}$
- mixed signed scalars $\{\psi_{ij}\}$; (non u.s.c. problem)
- piecewise affine functions $\{\phi_{ij}\} : \mathbb{R}^n \to \mathbb{R}$;
- $c : \mathbb{R}^n \to \mathbb{R}$ is a Bouligand differentiable function (locally Lipschitz and directional differentiable)
- a polyhedron set $P \subseteq \mathbb{R}^n$.

Affine Heaviside composite optimization problem (A-HSCOP)

Source problems:

- chance-constraint $\mathbb{P}\big(f(\mathbf{x},\xi) \geq 0\big) = \mathbb{E}\big[\mathbb{1}_{[0,\infty)}\big(f(\mathbf{x},\xi)\big)\big]$
- endogenous uncertainty $\mathbb{P}\big(\tilde{\xi} = \xi \mid \mathbf{x}\big) = \sum_{s=1}^{S} p_s \Big[\xi\, \mathbb{1}_{[0,\infty)}\big(\phi(\mathbf{x},\omega^s)\big) + (1-\xi)\,\mathbb{1}_{(-\infty,0)}\big(\phi(\mathbf{x},\omega^s)\big)\Big]$
- classification
- policy learning
- clustering

  . . .

Distinguished features:

- a mixed-signed combination of Heaviside composite functions, which leads to the loss of upper semicontinuity.
- piecewise affine inner functions.
- composition of the two highlighted features above.
- functional constraints with the same features further add to the challenges.
- large number of Heaviside functions: $O(N)$, $O(N^2)$

Solution methods for A-HSCOP

- **Discrete approach:** Given all affine coefficients are nonnegative,
A-HSCOP can be formulated as MIP with big-M technique
  - off-the-shelf solvers, BB methods for mid-size problems with univariate splits [Bertsimas & Dunn (2017)] [Verwer, Zhang (2019)] [Aghaei, Gómez, Vayanos (2025)]
  - often can not efficiently handle problems of large size, complicated by the piecewise affine inner functions and feasibility issue
  - can not apply to the A-HSCOP problem with mixed-signed coefficients.

Solution methods for A-HSCOP

PIP approach: in-between the continuous and discontinuous approach for A-HSCOP of a BIG size

- leverage state-of-art **MIP solver for mid-size problems progressively**, rather than solve the whole MIP problem
- take advantage of **progressively DC decomposition** for dealing with the piecewise affine inner functions
- guarantee the **iterative progressive improvement**, and characterize the local optimality to A-HSCOP, which is linked to global optimality through **controllable parameters**

A-HSCOP with nonnegative coefficients

Assume that:

- all affine coefficients $\{\psi_{ij}\}$ are nonnegative
- affine inner functions $\phi_{ij}(\mathbf{x})$
- there exists a constant $\underline{M}$ such that $\phi_{ij}(\mathbf{x}) \geq -M$ for all $\mathbf{x} \in P$

A-HSCOP $(\mathrm{P}^{\oplus}_{\mathrm{AHS}})$

$$\begin{aligned}\max_{\mathbf{x}\in P}\ & \theta^{\oplus}_{\mathrm{AHS}}(\mathbf{x}) \triangleq c(\mathbf{x}) \\ & + \sum_{j=1}^{J_0} \psi_{0j}\mathbb{1}_{[0,\infty)}\big(\phi_{0j}(\mathbf{x})\big) \\ \text{s.t. } & \sum_{j=1}^{J_i} \psi_{ij}\mathbb{1}_{[0,\infty)}\big(\phi_{ij}(\mathbf{x})\big) \geq b_i, \ \forall i\end{aligned}$$

MIP reformulation

$$\begin{aligned}\max_{\mathbf{x}\in P;\, \mathbf{z}}\ & c(\mathbf{x}) + \sum_{j=1}^{J_0} \psi_{0j} z_{0j} \\ \text{s.t. } & \sum_{j=1}^{J_i} \psi_{ij} z_{ij} \geq b_i, \ \forall i \\ & \phi_{ij}(\mathbf{x}) \geq -M(1 - z_{ij}), \ \forall j, i \\ & z_{ij} \in \{0, 1\}, \ \forall j, i.\end{aligned}$$

the number of Heaviside functions = the number of integer variables: $\sum_{i=0}^{m} J_i$.

Partial HSCOP and restricted MIP

With $\mathcal{J}^{\delta}_{i;>}(\bar{\mathbf{x}}) \subseteq \{j \in [J_i] \mid \phi_{ij}(\bar{\mathbf{x}}) > \delta_1\}$, $\mathcal{J}^{\delta}_{i;<}(\bar{\mathbf{x}}) \subseteq \{j \in [J_i] \mid \phi_{ij}(\bar{\mathbf{x}}) < -\delta_2\}$, $\mathcal{J}^{\delta}_{i;0}(\bar{\mathbf{x}}) \triangleq [J_i]/\big(\mathcal{J}^{\delta}_{i;>}(\bar{\mathbf{x}}) \cup \mathcal{J}^{\delta}_{i;<}(\bar{\mathbf{x}})\big)$,

partial A-HSCOP $(\mathrm{P}^{\oplus;\delta}_{\mathrm{AHS}}(\bar{\mathbf{x}}))$

$$\begin{aligned}
\max_{\mathbf{x}\in P}\ \theta^{\oplus}_{\mathrm{AHS}}(\mathbf{x};\bar{\mathbf{x}}) \triangleq c(\mathbf{x}) + \sum_{j\in\mathcal{J}^{\delta}_{0;0}(\bar{\mathbf{x}})} \psi_{0j}\mathbb{1}_{[0,\infty)}\big(\phi_{0j}(\mathbf{x})\big) + \sum_{j\in\mathcal{J}^{\delta}_{0;>}(\bar{\mathbf{x}})} \psi_{0j}\\
\text{s.t.}\ \sum_{j\in\mathcal{J}^{\delta}_{i;0}(\bar{\mathbf{x}})} \psi_{ij}\mathbb{1}_{[0,\infty)}\big(\phi_{ij}(\mathbf{x})\big) + \sum_{j\in\mathcal{J}^{\delta}_{i;>}(\bar{\mathbf{x}})} \psi_{ij} \geq \eta_i,\ \forall i\in[I]\\
\phi_{ij}(\mathbf{x}) \geq 0 \quad j\in\mathcal{J}^{+\delta}_{i;>}(\bar{\mathbf{x}}),\ \forall i\in[I].
\end{aligned}$$

restricted IP $(\mathrm{IP}^{\oplus;\delta}_{\mathrm{AHS}}(\bar{\mathbf{x}}))$

$$\begin{aligned}
\max_{\mathbf{x}\in P;\,\mathbf{z}}\ c(\mathbf{x}) + \sum_{j\in\mathcal{J}^{\delta}_{0;0}(\bar{\mathbf{x}})} \psi_{0j}z_{0j} + \sum_{j\in\mathcal{J}^{\delta}_{0;>}(\bar{\mathbf{x}})} \psi_{0j}\\
\text{s.t.}\ \sum_{j\in\mathcal{J}^{\delta}_{i;0}(\bar{\mathbf{x}})} \psi_{ij}z_{ij} + \sum_{j\in\mathcal{J}^{+\delta}_{i;>}(\bar{\mathbf{x}})} \psi_{ij} \geq b_i,\ \forall i\in[m]\\
\phi_{ij}(\mathbf{x}) \geq \underline{B}(1-z_{ij}),\ \forall (i,j)\in\mathcal{J}^{\delta}_{i;0}(\bar{\mathbf{x}})\\
\phi_{ij}(\mathbf{x}) \geq 0,\ (\Leftrightarrow z_{ij}=1)\ \forall (i,j)\in\mathcal{J}^{\delta}_{i;>}(\bar{\mathbf{x}})\\
z_{ij}\in\{0,1\},\ \forall (i,j)\in\mathcal{J}^{\delta}_{i;0}(\bar{\mathbf{x}}).
\end{aligned}$$

Computational Algorithm: PIP

Progressive Integer Programming (PIP) algorithm

At each iteration with feasible $\bar{\mathbf{x}}$,

• **MIP solver:** solve the restricted MIP problem at $\bar{\mathbf{x}}$ and obtain an optimal solution $\mathbf{x}_{\mathrm{new}}$ and corresponding optimal value $\theta^{\oplus}_{\mathrm{AHS}}(\mathbf{x}_{\mathrm{new}})$.

• **shrink:** if $\theta^{\oplus}_{\mathrm{AHS}}(\mathbf{x}_{\mathrm{new}}) > \theta^{\oplus}_{\mathrm{AHS}}(\bar{\mathbf{x}})$, shrink the in-between interval $[\delta_1, \delta_2]$;

• **expand:** if $\theta^{\oplus}_{\mathrm{AHS}}(\mathbf{x}_{\mathrm{new}}) = \theta^{\oplus}_{\mathrm{AHS}}(\bar{\mathbf{x}})$, expand the in-between interval $[\delta_1, \delta_2]$.

Iterative progressive improvement:

$$\theta^{\oplus}_{\mathrm{AHS}}(\mathbf{x}_{\mathrm{new}}) \geq \theta^{\oplus}_{\mathrm{AHS}}(\mathbf{x}_{\mathrm{new}}; \bar{\mathbf{x}}) \geq \theta^{\oplus}_{\mathrm{AHS}}(\bar{\mathbf{x}}; \bar{\mathbf{x}}) = \theta^{\oplus}_{\mathrm{AHS}}(\bar{\mathbf{x}})$$

the solution sequence progressively improves, and converges to a local maximizer.

Initialization

- It sometimes could be very hard to find a feasible solution satisfying a complicated constraint, such as the Gini constraint, or precision constraint.
- With challenging constraint, one can include an auxiliary variable as the residual of the constraint to guard against infeasibility.
- Specifically, we can apply the algorithms to the following problem:

$$\begin{aligned} \underset{\mathbf{x}\in P;\,\gamma\geq 0}{\textbf{maximize}} \quad & c(\mathbf{x}) + \sum_{j=1}^{J_0} \psi_{0j}\mathbb{1}_{[0,\infty)}\big(\phi_{0j}(\mathbf{x})\big) - \lambda\gamma \\ \textbf{subject to} \quad & \sum_{j=1}^{J_i} \psi_{ij}\mathbb{1}_{[0,\infty)}\big(\phi_{ij}(\mathbf{x})\big) + \gamma \geq \eta_i, \quad i = 1, \ldots, I; \end{aligned}$$

where the auxiliary variable γ is the constraint residual and $\lambda > 0$ is a penalty parameter (e.g., 10^4) whose aim is to enforce the feasibility condition $\gamma = 0$ if possible.

Initialization

- In practical implementation, one could start with an arbitrary solution $\mathbf{x}^0$ with a suitable constant γ_0 such that the constraints are satisfied. For instance, we could let $\mathbf{x}^0$ be an unconstrained minimizer of the A-HSCOP together with a suitable γ_0 value as the initial feasible solution.
- In the first several iterations of PIP, if IP solvers such as GUROBI produce a solution with $\gamma = 0$, then such a solution is also feasible to the original A-HSCOP problem, and we could proceed the PIP procedure without the γ variable afterwards.
- However, it is possible that in the algorithmic process of PIP, GUROBI may keep producing solutions with positive γ (infeasible solution) leading to the small progress with very small objective values. Possible reason is that for restricted IP subproblems with a smaller feasible region or even infeasible by fixing a portion of integer variables, it remains challenging for the IP solver to find a feasible solution.
- Therefore, one effective approach to deal with the above phenomenon is to reduce the size of the two fixing index sets $\mathcal{J}^{\delta}_{i;>}(\bar{\mathbf{x}})$ and $\mathcal{J}^{\delta}_{i;<}(\bar{\mathbf{x}})$ until a feasible solution is obtained.

In-between Index Sets and Update Rules

- In principle, after obtaining the new iterate point $\mathbf{x}^{\nu+1}$, the algorithm proceed with three updated disjoint index sets $\mathcal{J}^{\delta}_{i;>}(\mathbf{x}^{\nu+1})$, $\mathcal{J}^{\delta}_{i;<}(\mathbf{x}^{\nu+1})$, and $\mathcal{J}^{\delta}_{i;0}(\mathbf{x}^{\nu+1})$, such that the cardinality of the in-between index set $|\mathcal{J}^{\delta}_{i;0}(\mathbf{x}^{\nu+1})|$:
  - shrink when objective improves
  - expand when objective does not improve.
- How to construct in-between index set $\mathcal{J}^{\delta}_{i;0}(\mathbf{x}^{\nu+1})$, and how it shrinks or expands is very critical for practical efficiency. It may depend on the application contexts.

1. (quantile) control the ratio of the integer number to be contained in $\mathcal{J}^{\delta}_{i;0}(\bar{\mathbf{x}})$

$$\mathcal{J}^{\delta}_{i;0}(\bar{\mathbf{x}}) \triangleq \left\{ j \in [J_i] \mid -\delta_{2i} \leq \phi_{ij}(\bar{\mathbf{x}}) \leq \delta_{1i} \right\}$$

where δ_{1i}, δ_{2i} are r-upper and r-lower quantiles of $\{\phi_{ij}(\mathbf{x}^{\nu+1})\}_{j\in[J_i]}$ respectively

2. (softmax probability) construct the in-between index set via the softmax-based probabilities (for clustering problem, $\{\xi^s\}$ and $\{\bar{c}^k\}$ represent data points and the centroids)

$$\rho^{\tau}_{sk} \triangleq \frac{\exp\{-\|\xi^s - \bar{c}^k\|/\tau\}}{\sum_{k=1}^{K}\left(\exp\{-\|\xi^s - \bar{c}^k\|/\tau\}\right)}, \qquad \mathcal{J}^{\tau}_{0} \triangleq \left\{(s,k) : 1/K \leq \rho^{\tau}_{sk} < 1/2\right\}$$

r and τ are **expanded or shrinked** during PIP iterations.

In-between Index Sets and Update Rules

For controlling the ratio of the integer number, we take the following procedure:

- Compute $\mathbf{x}^{\nu+1}$ an optimal solution of $\left(\mathrm{P}^{\oplus;\delta}_{\mathrm{AHS}}(\mathbf{x}^{\nu})\right)$
  - If $\mathbf{x}^{\nu+1} = \mathbf{x}^{\nu}$, set $r_{\nu+1} \leftarrow \min\{r_{\nu} + r_{\Delta},\, r_{\max}\}$
  - If $\mathbf{x}^{\nu+1} \neq \mathbf{x}^{\nu}$, set $r_{\nu+1} \leftarrow \max\{r_{\nu} - r_{\Delta},\, r_{\min}\}$

  where $r_{\Delta}, r_{\max}, r_{\min} \in (0, 1)$ are predetermined scalars.
- Decompose the set of inner function values into two positive and non-positive subsets
$$\{\phi_{ij}(\mathbf{x}^{\nu+1})\}_{j\in[J_i]} = S^{+}_{i,\nu+1} \cup S^{-}_{i,\nu+1} = \{\phi^{+}_{ij}(\mathbf{x}^{\nu+1})\}_{j\in[J_i]} \cup \{\phi^{-}_{ij}(\mathbf{x}^{\nu+1})\}_{j\in[J_i]}$$
- Compute the lower and upper quantiles
$$\delta^{\nu+1}_{i;1} = \text{Lower\_quantile}\left(S^{+}_{i,\nu+1}; r_{\nu+1}\right), \quad \delta^{\nu+1}_{i;2} = \text{Upper\_quantile}\left(S^{-}_{i,\nu+1}; r_{\nu+1}\right)$$

Update in-between index set $\mathcal{J}^{\nu+1}_{i;0} \leftarrow \left\{j \mid \delta^{\nu+1}_{i;2} \leq \phi_{ij}(\mathbf{x}^{\nu+1}) \leq \delta^{\nu+1}_{i;1}\right\}$.

The aim of the update strategy is to enhance the performance of local optimality while saving the computational cost by the shrinkage or truncation strategy when needed.

Termination Criteria

- In theory, PIP requires to solve a sequence of partial Heaviside optimization problems of smaller sizes, which may be solved efficiently by IP solvers such as Gurobi.
- However, in practice, for the sake of computational efficiency, one could embed the early termination strategy when solving the subproblems in PIP; for instance, one could use callback mechanism to terminate the solver if the objective value remains unchanged for a period of time (e.g., 60s).
- With early termination of Gurobi in PIP, though the iterate point obtained is not guaranteed optimal, the algorithm still has the ascent property.
- With early termination of Gurobi in PIP, if at the terminating iteration, the obtained solution by Gurobi is optimal to the partial Heaviside problem, we can still claim that the obtained solution has local optimality regarding to the original Heaviside optimization problem.
- The whole PIP terminates when the objective does not improve for several consecutive expansions, yielding a local optimal solution. In practical implementation, the number of expansions is usually set heuristically depending on the time tolerance.

A-HSCOP with mixed-signed coefficients and PA $\phi_{ij}(\mathbf{x})$

Computational difficulties:

- $\psi_{ij} = \psi_{ij}^{+} - \psi_{ij}^{-}$ are mixed-signed ($\psi_{ij}^{+} \geq 0,\ \psi_{ij}^{-} \geq 0$), so the objective is not upper-semicontinuous and the constrained set is not closed.
- The inner function is a general PA $\phi_{ij}(\mathbf{x})$, that is the sum of a convex and a concave PA function:

$$\phi_{ij}(\mathbf{x}) = \max_{1 \leq k \leq K_{ij}} \left[(\mathbf{a}_{ij}^{k})^{\top} \mathbf{x} + \alpha_{ij}^{k} \right] + \min_{1 \leq \ell \leq L_{ij}} \left[(\mathbf{b}_{ij}^{\ell})^{\top} \mathbf{x} + \beta_{ij}^{\ell} \right]$$

A-HSCOP with mixed-signed coefficients and PA $\phi_{ij}(\mathbf{x})$

Approximation:

- With $\psi_{ij} < 0$, $\boxed{\psi_{ij}\mathbb{1}_{[0,\infty)}\big(\phi_{ij}(\mathbf{x})\big) \;\to\; \psi_{ij}\mathbb{1}_{(-\varepsilon,\infty)}\big(\phi_{ij}(\mathbf{x})\big)}$, which is a lower bound, and upper semicontinuous for some $\varepsilon > 0$

Decomposition:

- construct concave PA and convex PA functions with appropriate piece selection $(k,\ell) \in \mathcal{M}^\delta(\bar{\mathbf{x}})$ respectively for the usage depending on the sign of coefficients:

$$\phi_{ij}^{k_{ij};+}(\mathbf{x}) \triangleq (\mathbf{a}_{ij}^{k_{ij}})^\top \mathbf{x} + \alpha_{ij}^{k_{ij}} + \min_{1\le \ell' \le L_{ij}} \big[(\mathbf{b}_{ij}^{\ell'})^\top \mathbf{x} + \beta_{ij}^{\ell'}\big],$$

$$\phi_{ij}^{\ell_{ij};-}(\mathbf{x}) \triangleq \max_{1\le k' \le K_{ij}} \big[(\mathbf{a}_{ij}^{k'})^\top \mathbf{x} + \alpha_{ij}^{k'}\big] + (\mathbf{b}_{ij}^{\ell_{ij}})^\top \mathbf{x} + \beta_{ij}^{\ell_{ij}},$$

where

$$\mathcal{M}^\delta(\bar{\mathbf{x}}) = \prod_{i=0}^{I}\prod_{j=1}^{J_i}\Big\{\delta\text{-}\operatorname*{argmax}_{1\le k\le K_{ij}}\big[(\mathbf{a}_{ij}^k)^\top \bar{\mathbf{x}} + \alpha_{ij}^k\big]\Big\} \times \Big\{\delta\text{-}\operatorname*{argmin}_{1\le \ell\le L_{ij}}\big[(\mathbf{b}_{ij}^\ell)^\top \bar{\mathbf{x}} + \beta_{ij}^\ell\big]\Big\}.$$

A-HSCOP with mixed-signed coefficients and PA $\phi_{ij}(\mathbf{x})$

Decomposed ε-approximated subproblem $(\mathrm{P}^{\varepsilon;\mathbf{k},\ell}_{\mathrm{AHS}})(\bar{\mathbf{x}})$ u.s.c. problem

$$\begin{aligned}
\underset{\mathbf{x}\in P}{\textbf{maximize}}\ & V^{\varepsilon;k_0,\ell_0}(\mathbf{x}) \triangleq c(\mathbf{x}) + \sum_{j=1}^{J_0} \psi^+_{0j} \mathbb{1}_{[0,\infty)}\big(\phi^{k_{0j};+}_{0j}(\mathbf{x})\big) - \sum_{j=1}^{J_0} \psi^-_{0j} \mathbb{1}_{(-\varepsilon,\infty)}\big(\phi^{\ell_{0j};-}_{0j}(\mathbf{x})\big) \\
\textbf{subject to}\ & \sum_{j=1}^{J_i} \psi^+_{ij} \mathbb{1}_{[0,\infty)}\big(\phi^{k_{ij};+}_{ij}(\mathbf{x})\big) - \sum_{j=1}^{J_i} \psi^-_{ij} \mathbb{1}_{(-\varepsilon,\infty)}\big(\phi^{\ell_{ij};-}_{ij}(\mathbf{x})\big) \geq \eta_i,\ \forall i = 1, \ldots, I.
\end{aligned}$$

For all tuples $(\mathbf{k},\ell) = \big\{\{(k_{ij},\ell_{ij})\}_{j=1}^{J_i}\big\}_{i=0}^{I}$:

$$V(\mathbf{x}) \geq V^{\varepsilon;k_0,\ell_0}(\mathbf{x}) \quad \text{for any } \mathbf{x},$$

$$X^{\varepsilon';\mathbf{k},\ell}_{\mathrm{AHS}} \subseteq X^{\varepsilon;\mathbf{k},\ell}_{\mathrm{AHS}} \subseteq X_{\mathrm{AHS}}, \quad \forall \varepsilon' > \varepsilon > 0.$$

Computational Algorithm: IDSA-PIP

Iterative decomposed shrinkage algorithm (IDSA) – PIP

Initialization. Let $\{\varepsilon_\nu\}_{\nu=0}^{\infty}$ be a diminishing sequence. Let $\mathbf{x}^0$ be a feasible iterate to the problem $\left(\mathrm{P}_{\mathrm{AHS}}^{\varepsilon_0}\right)$, and set $\delta \in \mathbb{R}_+$.

For $\nu = 1, 2, \ldots$,

Main Computation. Solve the problem $\left(\mathrm{P}_{\mathrm{AHS}}^{\varepsilon_\nu;\mathbf{k},\ell}(\mathbf{x}^\nu)\right)$ via PIP algorithm for all pairs $(\mathbf{k}, \ell)$ in $\mathcal{M}^\delta(\mathbf{x}^\nu)$. Set $\mathbf{x}^{\nu+1}$ be the one that achieves the maximal objective value among all index pairs.

Terminate if $\mathbf{x}^{\nu+1}$ satisfies a prescribed rule (e.g., if $\|\mathbf{x}^{\nu+1} - \mathbf{x}^\nu\|$ is within a given tolerance).

Computational Algorithm: IDSA-PIP

To evaluate the effectiveness of decomposition and shrinkage techniques, we can have various version of PIP for comparisons:

1. **Full MIP**: with fixed ε, solve $\left(\mathrm{P}^{\varepsilon}_{\mathrm{AHS}}\right)$ by the IP solver GUROBI to its full MIP formulation with time limit.
2. **PIP**: with fixed ε, solve $\left(\mathrm{P}^{\varepsilon}_{\mathrm{AHS}}\right)$ by the PIP method (Algorithm 1), without the decomposition of inner PA functions.
3. **ISA-PIP**: A variant of the iterative shrinkage method, in which the inner loop computation is to apply PIP algorithm to $\left(\mathrm{P}^{\varepsilon_\nu}_{\mathrm{AHS}}\right)$, without the decomposition of the inner PA functions.
4. **IDSA-PIP**: A variant of the iterative shrinkage method, in which the inner loop computation is to apply the PIP method (Algorithm 1) to a decomposed approximating subproblem $\left(\mathrm{P}^{\varepsilon_\nu;\mathbf{k}_\nu,\ell_\nu}_{\mathrm{AHS}}\right)$ with an index pair $(\mathbf{k}_\nu, \ell_\nu)$ arbitrarily selected from the active index set for the inner PA functions.